

Web annex

Literature reviews
and evidence tables



The main document
is available at:

[https://apps.who.int/iris/bitstream/
handle/10665/370507/
9789240071193-eng.pdf](https://apps.who.int/iris/bitstream/handle/10665/370507/9789240071193-eng.pdf)

Package of interventions for

Rehabilitation

Module 5

Neurodevelopmental disorders



World Health
Organization

Package of interventions for rehabilitation. Module 5. Neurodevelopmental disorders. Web Annex. Literature reviews and evidence tables

(Package of interventions for rehabilitation)

ISBN 978-92-4-007121-6 (electronic version)

© World Health Organization 2023

Some rights reserved. This work is available under the Creative Commons Attribution-NonCommercial-ShareAlike 3.0 IGO licence (CC BY-NC-SA 3.0 IGO; <https://creativecommons.org/licenses/by-nc-sa/3.0/igo>).

Under the terms of this licence, you may copy, redistribute and adapt the work for non-commercial purposes, provided the work is appropriately cited, as indicated below. In any use of this work, there should be no suggestion that WHO endorses any specific organization, products or services. The use of the WHO logo is not permitted. If you adapt the work, then you must license your work under the same or equivalent Creative Commons licence. If you create a translation of this work, you should add the following disclaimer along with the suggested citation: “This translation was not created by the World Health Organization (WHO). WHO is not responsible for the content or accuracy of this translation. The original English edition shall be the binding and authentic edition”.

Any mediation relating to disputes arising under the licence shall be conducted in accordance with the mediation rules of the World Intellectual Property Organization (<http://www.wipo.int/amc/en/mediation/rules/>).

Suggested citation. Web Annex. Literature reviews and evidence tables. In: Package of interventions for rehabilitation. Module 5. Neurodevelopmental disorders. Geneva: World Health Organization; 2023 (Package of interventions for rehabilitation). Licence: CC BY-NC-SA 3.0 IGO.

Cataloguing-in-Publication (CIP) data. CIP data are available at <http://apps.who.int/iris>.

Sales, rights and licensing. To purchase WHO publications, see <https://www.who.int/publications/book-orders>. To submit requests for commercial use and queries on rights and licensing, see <https://www.who.int/copyright>.

Third-party materials. If you wish to reuse material from this work that is attributed to a third party, such as tables, figures or images, it is your responsibility to determine whether permission is needed for that reuse and to obtain permission from the copyright holder. The risk of claims resulting from infringement of any third-party-owned component in the work rests solely with the user.

General disclaimers. The designations employed and the presentation of the material in this publication do not imply the expression of any opinion whatsoever on the part of WHO concerning the legal status of any country, territory, city or area or of its authorities, or concerning the delimitation of its frontiers or boundaries. Dotted and dashed lines on maps represent approximate border lines for which there may not yet be full agreement.

The mention of specific companies or of certain manufacturers' products does not imply that they are endorsed or recommended by WHO in preference to others of a similar nature that are not mentioned. Errors and omissions excepted, the names of proprietary products are distinguished by initial capital letters.

All reasonable precautions have been taken by WHO to verify the information contained in this publication. However, the published material is being distributed without warranty of any kind, either expressed or implied. The responsibility for the interpretation and use of the material lies with the reader. In no event shall WHO be liable for damages arising from its use.

This publication forms part of the document entitled *Package of interventions for rehabilitation. Module 5. Neurodevelopmental disorders*. It is being made publicly available for transparency purposes and information.

Contents

Introduction	1
References	1
1. Package of interventions for rehabilitation for autism spectrum disorders	2
1.1. Literature search for clinical practice guidelines for rehabilitation in autism spectrum disorders	2
1.2. Selection of clinical practice guidelines for autism spectrum disorders	3
1.3. Cochrane systematic reviews included for autism spectrum disorders	7
1.4. Evidence tables for autism spectrum disorders	8
1.5. References	20
2. Package of interventions for rehabilitation for disorders of intellectual development	22
2.1. Literature search for clinical practice guidelines for rehabilitation in disorders of intellectual development	22
2.2. Selection of clinical practice guidelines for disorders of intellectual development	23
2.3. Cochrane systematic reviews included for disorders of intellectual development	27
2.4. Evidence tables for disorders of intellectual development	27
2.5. References	41

Introduction

This Web Annex provides information on the literature search and selection of the clinical practice guidelines and Cochrane systematic reviews included in the *Package of interventions for rehabilitation. Module 5. Neurodevelopmental disorders (1)*. The methods applied for the search and selection are described in the Introduction to the Package of interventions for rehabilitation (2). The Web Annex further provides background information on the selected guidelines and evidence tables for the interventions included in the health condition specific packages.

References

1. Package of interventions for rehabilitation. Module 5. Neurodevelopmental disorders. Geneva: World Health Organization; 2023 (Package of interventions for rehabilitation). Licence: CC BY-NC-SA 3.0 IGO.
2. Package of interventions for rehabilitation. Module 1. Introduction. Geneva: World Health Organization; 2023 (Package of interventions for rehabilitation). Licence: CC BY-NC-SA 3.0 IGO.

1. Package of interventions for rehabilitation for autism spectrum disorders

1.1. Literature search for clinical practice guidelines for rehabilitation in autism spectrum disorders

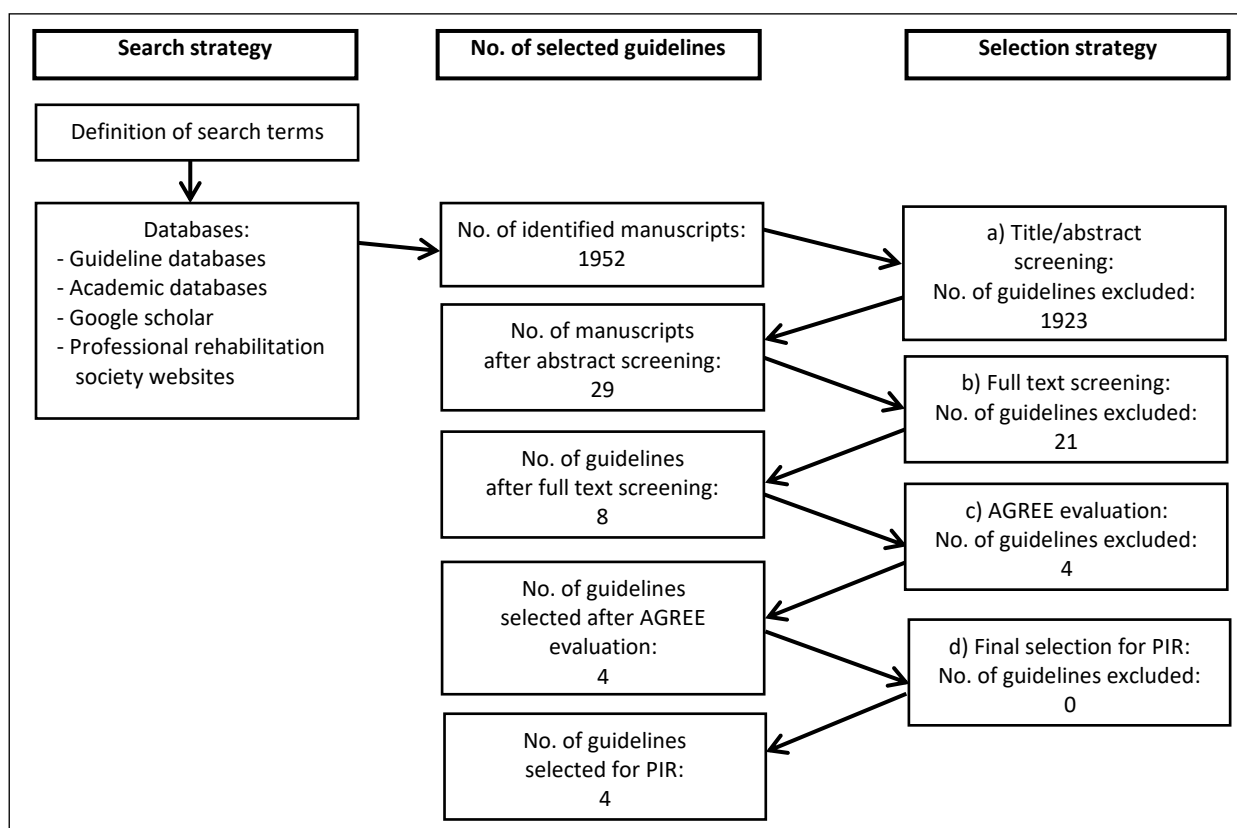
Table D1.1. Searched academic databases and search terms

Academic databases	Search terms
PubMed	(autism (tiab) OR autistic(tiab) OR autism spectrum disorder(mesh) OR “pervasive developmental disorder” (tiab)) AND (guideline(tiab) OR guidelines(tiab) OR guideline (publication type))
EMBASE	(autism:ti,ab OR autistic:ti,ab OR 'autism'/exp/mj OR 'autism spectrum disorder':ti,ab OR 'pervasive developmental disorder':ti,ab) AND (guideline:ti,ab OR guidelines:ti,ab) AND (english)/lim AND (2009-2019)/py
CINAHL PLUS	((MM "Autistic Disorder") OR autism OR autistic OR pervasive developmental disorder) AND TI (guideline OR guidelines)
PEDro	Abstract & Title: Autis*
SCOPUS	(TITLE-ABS-KEY (autism OR autistic OR "pervasive developmental disorder") AND TITLE-ABS-KEY (guideline OR guidelines)) AND (LIMIT-TO (PUBYEAR , 2019) OR LIMIT-TO (PUBYEAR , 2018) OR LIMIT-TO (PUBYEAR , 2017) OR LIMIT-TO (PUBYEAR , 2016) OR LIMIT-TO (PUBYEAR , 2015) OR LIMIT-TO (PUBYEAR , 2014) OR LIMIT-TO (PUBYEAR , 2013) OR LIMIT-TO (PUBYEAR , 2012) OR LIMIT-TO (PUBYEAR , 2011) OR LIMIT-TO (PUBYEAR , 2010) OR LIMIT-TO (PUBYEAR , 2009)) AND (LIMIT-TO (LANGUAGE , "English")))
Web of science	(TS=(autism OR autistic OR "pervasive developmental disorder") AND TS=(guideline OR guidelines)) AND LANGUAGE: (English)

Table D1.2. Searched non-academic databases

Guideline databases
• Guidelines International Network (GIN) • National Institute for Health and Care Excellence (NICE, United Kingdom) • National Health and Medical Research Council (NHMRC, Australia) • Scottish Intercollegiate Guidelines Network (SIGN) • Canadian Medical Association (CMA) Infobase: Clinical Practice Guidelines • New Zealand Guidelines Group (NZGG) • National Guideline Clearinghouse (NGC, United States) • eGuidelines
Professional rehabilitation society websites
• American Academy of Neurology • American Academy of Pediatrics • American Academy of Child & Adolescent Psychiatry • American Psychiatric Association • American Occupational Therapy Association • Society for Developmental & Behavioral Pediatrics • American Speech-Language-Hearing Association

Fig. D1.1. Flow chart for the search and selection of clinical practice guidelines for rehabilitation for autism spectrum disorders



AGREE: Appraisal of Guidelines for Research and Evaluation; No: number; PIR: package of interventions for rehabilitation.

1.2. Selection of clinical practice guidelines for autism spectrum disorders

Table D1.3. Results of the AGREE^a evaluation

Guidelines included for AGREE evaluation	Average of key items ^b				Average of summary value of items 4, 7, 8, 10, 12, 13, 15, 22, 23 ^b	Selection criteria fulfilled
	7	8	12	22		
Belgian Health Care Knowledge Centre: Management of autism in children and young people: a good clinical practice guideline. 2014 (1).	7	6	6	7	58	Yes
National Institute for Health and Care Excellence (NICE): The NICE guideline on recognition, referral, diagnosis, and management of adults on the autism spectrum. NICE Guideline 142. 2012 (2).	7	6	7	6	56	Yes
National Institute for Health and Care Excellence (NICE): The NICE guideline on the management and support of children and young people on the autism spectrum. NICE Guideline 170. 2013 (3).	7	6	7	5	54	Yes
Scottish Intercollegiate Guidelines Network (SIGN): Assessment, diagnosis, and interventions for autism spectrum disorders. SIGN 145. 2016 (4).	6	3	5	6	46	Yes
Ministries of Health and Education: New Zealand autism spectrum disorder guideline (2 nd edition). 2016 (5).	7	7	4	3	43.5	No
Malaysian Health Technology Assessment Section (MaHTAS): Management of autism spectrum disorder in children and adolescents. 2018 (6).	5	3	5	4	35.33	No

Guidelines included for AGREE evaluation	Average of key items ^b				Average of summary value of items 4, 7, 8, 10, 12, 13, 15, 22, 23 ^b	Selection criteria fulfilled
	7	8	12	22		
New York State Department of Health: Clinical practice guideline on assessment and intervention services for young children with autism spectrum disorders (ASD). Update 2017 (7).	5	5	3	2	31.33	No
National Autism Centre: Findings and Conclusions: National Standards Project, Phase 2—Addressing the need for evidence-based practice guidelines for autism spectrum disorder. 2015 (8).	5	6	2	1	23.67	No

^a AGREE (9).

^b Numbers highlighted in **bold** in the table indicate a criterion that has not been fulfilled.

Table D1.4. Final selection of clinical practice guidelines

Guidelines included after AGREE evaluation	Selection criteria			Finally selected
	Publication date	Comprehensive	Multi-professional	
Belgian Health Care Knowledge Centre: Management of autism in children and young people: a good clinical practice guideline. 2014 (1).	2014	Yes	Yes	Yes
National Institute for Health and Care Excellence (NICE): The NICE guideline on recognition, referral, diagnosis, and management of adults on the autism spectrum. NICE Guideline 142. 2012 (2).	2012	Yes	Yes	Yes
National Institute for Health and Care Excellence (NICE): The NICE guideline on the management and support of children and young people on the autism spectrum. NICE Guideline 170. 2013 (3).	2013	Yes	Yes	Yes
Scottish Intercollegiate Guidelines Network (SIGN): Assessment, diagnosis, and interventions for autism spectrum disorders. SIGN 145. 2016 (4).	2016	Yes	Yes	Yes

Additional WHO guideline on mental health interventions

mhGAP Intervention guide for mental, neurological and substance use disorders in non-specialized health settings: mental health GAP Action Programme (mhGAP) – version 2.0. 2016 (10).

Table D1.5. Information on the included clinical practice guidelines

Guidelines included	Abbreviation ^a	Target population	Topic of the guideline
Belgian Health Care Knowledge Centre: Management of Autism in Children and Young People: A Good Clinical Practice Guideline. 2014 (1).	KCE	Children and adolescents (0-18 years) with ASD, their family, and carers.	Recommendations for treatment and support of children and adolescents with ASD and their family
National Institute for Health and Care Excellence (NICE): The NICE guideline on recognition, referral, diagnosis, and management of adults on the autism spectrum. 2012 (2).	NICE1	Adults (18 or older), with suspected or diagnosed high functioning or low functioning autistic spectrum conditions.	Recommendations for the support and management of adults with ASD

Guidelines included	Abbreviation^a	Target population	Topic of the guideline
National Institute for Health and Care Excellence (NICE): The NICE guideline on the management and support of children and young people on the autism spectrum. 2013 (3).	NICE2	Children and young people on the autism spectrum.	Recommendations for the management and support of children and young people with ASD
Scottish Intercollegiate Guidelines Network (SIGN): Assessment, diagnosis, and interventions for autism spectrum disorders. 2016 (4).	SIGN	Children, young people, adults, and older adults with ASD	Recommendations for best practice in the assessment, diagnosis, and interventions for people with ASD.
mhGAP Intervention guide for mental, neurological and substance use disorders in non-specialized health settings: mental health GAP Action Programme (mhGAP) – version 2.0. 2016 (10).	mhGAP	People with mental, neurological, and substance use disorders	mhGAP-CHILD: mental health interventions for children and adolescents with mental and behavioural disorders mhGAP-DEP: mental health interventions for depression mhGAP-OTH: mental health interventions for other significant emotional and medical unexplained somatic complaints.

^a Abbreviated titles are used in the evidence tables as references to the corresponding guideline.

Table D1.6. Classification of the strength of recommendation in the included clinical practice guidelines

Clinical practice guideline	Strength of recommendation		
	Strong	Weak/Conditional	Expert opinion
KCE (1)	Strong: The clinical practice guideline used the "should" to formulate the interventions. The WHO team interpreted this word as "strong".	Conditional: The clinical practice guideline used the "consider" to formulate the interventions. The WHO team interpreted this word as "conditional".	When no grade could be allocated, working group members attempted to reach professional agreement (PA) on specific topics.
NICE1 (2) NICE2 (3)	Strong in favour: Recommendations on interventions that 'should' be used, the guideline development group is confident that, for the vast majority of people, the intervention (or interventions) will do more good than harm.	Conditional: Recommendations on interventions that should be 'considered', the guideline development group is confident that the intervention will do more good than harm for most patients. The choice of intervention is therefore more likely to vary depending on a person's values and preferences, and so the healthcare professional should spend more time discussing the options with the patient.	-
SIGN (4)	Strong in favour: Recommendations on interventions that 'should' be used, the guideline development group is confident that, for the vast majority of people, the intervention (or interventions) will do more good than harm.	Conditional: Recommendations on interventions that should be 'considered', the guideline development group is confident that the intervention will do more good than harm for most patients. The choice of intervention is therefore more likely to vary depending on a person's values and preferences, and so the healthcare professional should spend more time discussing the options with the patient.	Good practice points: Recommended best practice based on the clinical experience of the guideline development group.
mhGAP (10)	Strong: Recommendation suggests that the GDG agreed that the quality of the evidence combined with certainty about the values and preferences and the feasibility of the recommendation meant it should be followed in all or almost all circumstances.	Conditional: Recommendation suggests less certainty about the quality of evidence and variation values and preferences and feasibility, leading to circumstances in which the recommendation may not apply.	Good practice points: Recommended best practice based on the clinical experience of the guideline development group.

KCE: Belgian Health Care Knowledge Centre (1); mhGAP: WHO Mental Health Gap Action Programme Intervention Guide (10); NICE1: National Institute for Health and Care Excellence (2); NICE2: National Institute for Health and Care Excellence (3); SIGN: Scottish Intercollegiate Guidelines Network (4).

Table D1.7. Classification of the quality of the evidence in the included clinical practice guidelines

Clinical practice guideline	Quality of the evidence			
	High	Moderate	Low	Very low
KCE (1)	-	-	Low: The final KCE recommendations for management and treatment of ASD are consensus based and therefore their level of evidence is low.	-
NICE1 (2) NICE2 (3)	High: High confidence in the correlation between true and estimated effect	Moderate: Moderate confidence in the estimated effect. It is possible that the true effect is very different from the estimated effect.	Low: Limited confidence in the estimated effect. The true effect may be very different from the estimated effect.	Very low: Very little confidence in the estimated effect. The true effect is very probably different from the estimated effect.
SIGN (4)	1++: High quality meta-analyses, systematic reviews of randomized controlled trials (RCTs), or RCTs with a very low risk of bias. 1+: Well conducted meta-analyses, systematic reviews, or RCTs with a low risk of bias.	1-: Meta-analyses, systematic reviews, or RCTs with a high risk of bias. 2++: High quality systematic reviews of case control or cohort studies, High quality case control or cohort studies with a very low risk of confounding or bias and a high probability that the relationship is causal.	2+: Well conducted case control or cohort studies with a low risk of confounding or bias and a moderate probability that the relationship is causal. 2-: Case control or cohort studies with a high risk of confounding or bias and a significant risk that the relationship is not causal.	3: Non-analytic studies, e.g. case reports, case series. 4: Expert opinion.
mhGAP (10)	High: High confidence in the correlation between true and estimated effect.	Moderate: Moderate confidence in the estimated effect. It is possible that the true effect is very different from the estimated effect.	Low: Limited confidence in the estimated effect. The true effect may be very different from the estimated effect.	Very low: Very little confidence in the estimated effect. The true effect is very probably different from the estimated effect.

KCE: Belgian Health Care Knowledge Centre (1); mhGAP: WHO Mental Health Gap Action Programme Intervention Guide (10); NICE1: National Institute for Health and Care Excellence (2); NICE2: National Institute for Health and Care Excellence (3); RCTs: randomized controlled trials; SIGN: Scottish Intercollegiate Guidelines Network (4).

1.3. Cochrane systematic reviews included for autism spectrum disorders

Table D1.8. Cochrane systematic reviews included for autism spectrum disorders

Author, title
• Brignell A, Chenausky K, Song H, Zhu J, Suo C, Morgan A. Communication interventions for autism spectrum disorder in minimally verbal children. 2018 (11). • Reichow B, Hume K, Barton E, Boyd B. Early intensive behavioral intervention (EIBI) for young children with autism spectrum disorders (ASD). 2018 (12). • Reichow B, Steiner A, Volkmar F. Social skills groups for people aged 6 to 21 with autism spectrum disorders (ASD). 2012 (13).

1.4. Evidence tables for autism spectrum disorders

Table D1.9. Evidence table for assessments and interventions included in the *Package of interventions for rehabilitation for autism spectrum disorders*

Assessments and interventions	Original recommendation from clinical practice guidelines/Development group members' rationales	Strength of recommendation ^a	Quality of the evidence ^b	Reference ^c
Cognitive functions				
Assessment of cognitive functions	Individuals with autism spectrum disorders (ASD) should be considered for assessment of intellectual, neuropsychological, and adaptive functioning.	Strong	3, 4	SIGN (4): p. 16; 4.3.2
	The assessment should make best use of existing documentation about personal, educational, occupational, social and communication functioning, and should include assessment of any coexisting conditions, especially depression, anxiety, attention deficit hyperactivity disorder, obsessive-compulsive disorder and global delay or intellectual disability in line with autism in adults (NICE clinical guideline 142).	Strong	n.a.	NICE2 (3): p. 179; Sec 5.5.1
Provision and training in the use of assistive products for cognition	Cognitive impairments are frequent in people with ASD. The development group considers the provision and training in the use of assistive products for cognitive functions as an effective intervention to compensate cognitive impairments. To the knowledge of the development group, no severe harms are associated with the provision of assistive products. The cost-benefit ratio in relation to the clinical benefits justifies the inclusion of the assessment in the Package of interventions for rehabilitation. Furthermore, the provision of assistive products is considered as feasible and accepted by people with ASD.	n.a.	n.a.	DG expert opinion
Sleep functions				
Assessment of sleep disturbances	If the child or young person with autism snores loudly, chokes or appears to stop breathing while sleeping, refer to a specialist to check for obstructive sleep apnoea.	Strong	n.a.	NICE2 (3): p. 618; 8.8.9
	It is recommended to first offer a detailed clinical assessment in children with autism and sleep problems.	Strong	Low	KCE (1): p. 103; Tab. 18
	If a child or young person with autism develops a sleep problem offer an assessment that identifies: • what the sleep problem is (for example, delay in falling asleep, frequent waking, unusual behaviours, breathing problems or sleepiness during the day) • day and night sleep patterns, and any change to those patterns • whether bedtime is regular • what the sleep environment is like for example: – the level of background noise – use of a blackout blind – a television or computer in the bedroom – whether the child shares the room with someone • presence of comorbidities especially those that feature hyperactivity or other behavioural problems • levels of activity and exercise during the day • possible physical illness or discomfort (for example, reflux, ear or tooth ache, constipation or eczema) • effects of any medication • any other individual factors thought to enhance or disturb sleep, such as emotional relationships or problems at school • the impact of sleep and behavioural problems on parents or carers and other family members.	Strong	n.a.	NICE2 (3): p. 617; 8.8.9
Behavioural interventions	Develop a sleep plan (this will often be a specific sleep behavioural intervention) with the parents or carers to help address the identified sleep problems and to establish a regular night-time sleep pattern. Ask the parents or carers to record the child or young person's sleep and wakefulness throughout the day and night over a 2-week period. Use this information to modify the sleep plan if necessary and review the plan regularly until a regular sleep pattern is established.	Strong	n.a.	NICE2 (3): p. 618; 8.8.9
Problems with behaviours (challenging behaviours)				
	Anger management interventions should typically include: • functional analysis of anger and anger-provoking situations • coping-skills training and behaviour rehearsal • relaxation training • development of problem-solving skills.	Strong	n.a.	NICE1 (2): p. 217; 7.6.7

Assessment of problems with behaviour	Psychosocial interventions for behaviour that challenges should include: • clearly identified target behaviour • a focus on outcomes that are linked to quality of life • assessment and modification of environmental factors that may contribute to initiating or maintaining the behaviour • a clearly defined intervention strategy that takes into account the developmental level and coexisting problems of the child or young person • a specified timescale to meet intervention goals (to promote modification of intervention strategies that do not lead to change within a specified time) • a systematic measure of the target behaviour taken before and after the intervention to ascertain whether the agreed outcomes are being met • consistent application in all areas of the child or young person's environment (for example, at home and at school) • agreement among parents, carers and professionals in all settings about how to implement the intervention.	Strong	n.a.	NICE2 (3): p. 439; 7.7.1
	In the case of challenging behaviour, a functional assessment should first be performed.	Strong	Expert consensus	KCE (1): p. 99; Tab. 18
	The functional assessment should identify: • triggers for the behaviour, • patterns of behaviour, • the needs that the child or young person is attempting to meet by performing the behaviour, • the consequences of the behaviour.	Strong	Expert consensus	KCE (1): p. 50; Tab. 5; p. 99; Tab. 18
	When assessing challenging behaviour carry out a functional analysis (see recommendation 5.4.7.21) including identifying and evaluating any factors that may trigger or maintain the behaviour, such as: • physical disorders • the social environment (including relationships with family members, partners, carers and friends) • the physical environment, including sensory factors • coexisting mental disorders (including depression, anxiety disorders and psychosis) • communication problems • changes to routines or personal circumstances.	Strong	Not reported	NICE1 (2): p. 141; 5.4.7
	Assess factors that may increase the risk of behaviour that challenges in routine assessment and care planning in children and young people with autism, including: - impairments in communication that may result in difficulty understanding situations or in expressing needs and wishes - coexisting physical disorders, such as pain or gastrointestinal disorders - coexisting mental health problems such as anxiety or depression and other neurodevelopmental conditions such as ADHD - the physical environment, such as lighting and noise levels - the social environment, including home, school and leisure activities - changes to routines or personal circumstances - developmental change, including puberty - exploitation or abuse by others - inadvertent reinforcement of behaviour that challenges - the absence of predictability and structure.	Strong	Not reported	NICE2 (3): p. 438; Sec 7.7.1
Behavioural interventions	Psychosocial interventions for behaviour that challenges should include assessment and modification of environmental factors that may contribute to initiating or maintaining the behaviour.	Strong	Expert consensus	KCE (1): p. 100; Tab. 18
	Behavioural interventions for children and adolescents, and caregiver skills training, may be offered for the treatment of behavioural disorders. Behavioural interventions (behavioural and cognitive behavioural therapies, school-based therapies, and caregiver skills training).	Conditional	Low	mhGAP (10): p. 39
	Offer a psychosocial intervention for the challenging behaviour first if no coexisting mental or physical disorder, or problem related to the physical or social environment, has been identified as triggering or maintaining challenging behaviour: Cognitive-behavioural interventions.	Strong	very low to low.	NICE1 (2): p. 142; 5.4.7
	Psychosocial interventions for behaviour that challenges should include: • clearly identified target behaviour • a focus on outcomes that are linked to quality of life • assessment and modification of environmental factors that may contribute to initiating or maintaining the behaviour • a clearly defined intervention strategy that takes into account the developmental level and coexisting problems of the child or young person • a specified timescale to meet intervention goals (to promote modification of intervention strategies that do not lead to change within a specified time) • a systematic measure of the target behaviour taken before and after the intervention to ascertain whether the agreed outcomes are being met • consistent application in all areas of the child or young person's environment (for example, at home and at school) • agreement among parents, carers and professionals in all settings about how to implement the intervention.	Strong	n.a.	NICE2 (3): p. 439; 7.7.1

	Psychosocial interventions for challenging behaviour should include: • clearly identified target behaviour(s) • a focus on outcomes that are linked to quality of life • assessment and modification of environmental factors that may contribute to initiating or maintaining the behaviour • a clearly defined intervention strategy • a clear schedule of reinforcement, and capacity to offer reinforcement promptly and contingently on demonstration of the desired behaviour • a specified timescale to meet intervention goals (to promote modification of intervention strategies that do not lead to change within a specified time) • a systematic measure of the target behaviour(s) taken before and after the intervention to ascertain whether the agreed outcomes are being met.	Strong	n.a.	NICE1 (2): p. 207; 7.5.7
	Offer the following to address factors that may trigger or maintain behaviour that challenges: - treatment for physical disorders or coexisting mental health and behavioural problems - interventions aimed at changing the environment, such as: - providing advice to families and carers - making adjustments or adaptations to the physical surroundings.	Strong	n.a.	NICE2 (3): p. 438; 7.7.1
	Psychosocial interventions for behaviour that challenges should include assessment and modification of environmental factors that may contribute to initiating or maintaining the behaviour.	Strong	Expert consensus	KCE (1): p. 100; Tab. 18
	Anger management interventions should typically include: • functional analysis of anger and anger-provoking situations • coping-skills training and behaviour rehearsal • relaxation training • development of problem-solving skills.	Strong	n.a.	NICE1 (2): p. 217; 7.6.7
Physical exercise training	Challenging and stereotyped behaviours are often present in people with ASD. The development group considers physical exercise training as effective to improve such problems with behaviour in people with ASD. To the knowledge of the development group, no severe harms are associated with performing physical exercise training in people with ASD. The cost-benefit ratio in relation to the clinical benefits justifies the inclusion of this intervention in the Package of interventions for rehabilitation. Furthermore, physical exercise training is considered as feasible and accepted by people with ASD. This rationale is based on the clinical expertise of the development group members and available evidence on the effectiveness of physical exercise training in people with ASD (14-16).	n.a.	n.a.	DG expert opinion
Relaxation training	Anger management interventions should typically include: • functional analysis of anger and anger-provoking situations • coping-skills training and behaviour rehearsal • relaxation training • development of problem-solving skills.	Strong	n.a.	NICE1 (2): p. 217; 7.6.7
Perceptual functions				
Assessment of perceptual functions	People with ASD often experience problems with sensory perception and processing. The development group considers the assessment of sensory perception and processing challenges as essential for the determination of intervention needs. To the knowledge of the development group, no severe harms are associated with the assessment. The cost-benefit ratio in relation to the clinical benefits justifies the inclusion of the assessment in the Package of interventions for rehabilitation. Furthermore, the assessment is considered as feasible and accepted by people with ASD.	n.a.	n.a.	DG expert opinion
Sensory integration interventions	People with ASD often experience problems with sensory perception and processing. The development group considers sensory integration interventions as effective to reduce problems with sensory functions in people with ASD. To the knowledge of the development group, no severe harms are associated with applying sensory interventions in people with ASD. The cost-benefit ratio in relation to the clinical benefits justifies the inclusion of this intervention in the Package of interventions for rehabilitation. Furthermore, sensory integration interventions are considered as feasible and accepted by people with ASD. This rationale is based on the clinical expertise of the development group members and available evidence on the effectiveness of sensory integration interventions in people with ASD (17-19).	n.a.	n.a.	DG expert opinion
Provision and training in the use of assistive products for perceptual functions	People with ASD often experience problems with sensory perception and processing. The development group considers the provision and training in the use of assistive products for sensory perception challenges as an effective intervention to address these limitations. To the knowledge of the development group, no severe harms are associated with the provision of assistive products. The cost-benefit ratio in relation to the clinical benefits justifies the inclusion of the assessment in	n.n.	n.a.	DG expert opinion

	the Package of interventions for rehabilitation. Furthermore, the provision of assistive products is considered as feasible and accepted by people with ASD.			
Cognitive functions of language				
Assessment of language	All children and young people with ASD should have a comprehensive evaluation of their speech and language and communication skills, which should inform intervention.	Strong	3	SIGN (4): p. 16; 4.3.1
Language therapy	Based on expert consensus speech therapy is recommended in autistic children with identified speech and language problems.	Strong	Expert consensus	KCE (1): p. 102; Tab. 18
Speech functions				
Assessment of speech functions	All children and young people with ASD should have a comprehensive evaluation of their speech and language and communication skills, which should inform intervention.	Strong	3	SIGN (4): p. 16; 4.3.1
Speech therapy	Based on expert consensus speech therapy is recommended in autistic children with identified speech and language problems.	Strong	Expert consensus	KCE (1): p. 102; Tab. 18
Communication				
Assessment of communication	All children and young people with ASD should have a comprehensive evaluation of their speech and language and communication skills, which should inform intervention.	Strong	3	SIGN (4): p. 16; 4.3.1
	Healthcare professionals involved in specialist assessment should take an ASD-specific developmental history and should directly observe and assess the individual's social and communication skills and behaviour.	Strong	2+, 4	SIGN (4): p. 15; 4.2.3
	The assessment should make best use of existing documentation about personal, educational, occupational, social and communication functioning, and should include assessment of any coexisting conditions, especially depression, anxiety, attention deficit hyperactivity disorder, obsessive-compulsive disorder and global delay or intellectual disability in line with Autism in adults (NICE clinical guideline 142).	Strong	n.a.	NICE2 (3): p. 179; 5.5.1
	When assessing challenging behaviour carry out a functional analysis (see recommendation 5.4.7.21) including identifying and evaluating any factors that may trigger or maintain the behaviour, such as: ● physical disorders ● the social environment (including relationships with family members, partners, carers and friends) ● the physical environment, including sensory factors ● coexisting mental disorders (including depression, anxiety disorders and psychosis) ● communication problems ● changes to routines or personal circumstances.	Strong	n.a.	NICE1 (2): p. 141; 5.4.7
Communication skills training	96 Based on expert consensus speech and language problems in children with autism should be addressed within a personalized project including functional objectives in the field of verbal or non-verbal communication. This programme could include augmentative communication, such as Picture Exchange Communication System (PECS) and should be initiated early on.	Strong	Expert consensus	KCE (1): p. 102; Tab. 18
	Access to support from staff trained in applied behaviour analysis-based technologies (PECS), discrete trial training, task analysis, prompting, fading or shaping) to build independence in adaptive, communication and social skills should be considered for children with ASD.	Conditional	1++, 2++, 2+	SIGN (4): p. 24; 6.3.1
Provision and training in the use of assistive products for communication	Interventions to support communicative understanding and expression in individuals with ASD, such as the PECS and the use of environmental visual supports (e.g. in the form of pictures or objects), should be considered.	Conditional	1++, 1-, 3,	SIGN (4): p. 22; 6.2.1
	Access to support from staff trained in applied behaviour analysis-based technologies (PECS), discrete trial training, task analysis, prompting, fading or shaping) to build independence in adaptive, communication and social skills should be considered for children with ASD.	Conditional	1++, 2++, 2+	SIGN (4): p. 24; 6.3.1
	Based on expert consensus the guideline development group provided a recommendation to use augmentative communication (such as PECS) for children with autism and impairment in adaptive behaviour.	Strong	Expert consensus	KCE (1): p. 101; Tab. 18

	Based on expert consensus speech and language problems in children with autism should be addressed within a personalized project including functional objectives in the field of verbal or non-verbal communication. This programme could include augmentative communication, such as PECS and should be initiated early on.	Strong	Expert consensus	KCE (1): p. 102; Tab. 18
Nutrition				
Assessment of nutritional status	People with ASD are at higher risk for malnutrition. The development group considers the assessment of nutritional status as essential for the determination of intervention needs. To the knowledge of the development group, no severe harms are associated with the assessment. The cost-benefit ratio in relation to the clinical benefits justifies the inclusion of the assessment in the Package of interventions for rehabilitation. Furthermore, the assessment is considered as feasible and accepted by people with ASD (20).	n.a.	n.a.	DG expert opinion
Behavioural interventions	People with ASD often show specific behaviours regarding food selectivity and sensitivity which increase the risk for malnutrition. The development group considers behavioural interventions as effective in addressing food selectivity and sensitivity in people with ASD. To the knowledge of the development group, no severe harms are associated with the use of behavioural interventions in ASD. The cost-benefit ratio in relation to the clinical benefits justifies the inclusion of this intervention in the Package of interventions for rehabilitation. Furthermore, behavioural interventions are considered as feasible and accepted by people with ASD. This rationale is based on the clinical expertise of the development group members and available evidence on the effectiveness of behavioural interventions in people with ASD (21).	n.a.	n.a.	DG expert opinion
Assessment of food selectivity and sensitivity	People with ASD often show specific behaviours regarding food selectivity and sensitivity which increase the risk for malnutrition. The development group considers the assessment of food sensitivity and selectivity as essential for the determination of intervention needs. To the knowledge of the development group, no severe harms are associated with the assessment. The cost-benefit ratio in relation to the clinical benefits justifies the inclusion of the assessment in the Package of interventions for rehabilitation. Furthermore, the assessment is considered as feasible and accepted by people with ASD (22).	n.a.	n.a.	DG expert opinion
Sensory stimulation	People with ASD often show specific behaviours regarding food selectivity and sensitivity which increase the risk for malnutrition. The development group considers sensory stimulation as effective in addressing food selectivity and sensitivity in people with ASD. To the knowledge of the development group, no severe harms are associated with the use of sensory stimulation in ASD. The cost-benefit ratio in relation to the clinical benefits justifies the inclusion of this intervention in the Package of interventions for rehabilitation. Furthermore, sensory stimulation is considered as feasible and accepted by people with ASD. This rationale is based on the clinical expertise of the development group members and available evidence on the effectiveness of sensory interventions in people with ASD (21).	n.a.	n.a.	DG expert opinion
Sexual functions and intimate relationships				
Assessment of sexual functions and intimate relationships	People with ASD often experience challenges with sexual functions and intimate relationships. The development group considers the assessment of sexual functions and intimate relationships as essential for the determination of intervention needs. To the knowledge of the development group, no severe harms are associated with the assessment. The cost-benefit ratio in relation to the clinical benefits justifies the inclusion of the assessment in the Package of interventions for rehabilitation. Furthermore, the assessment is considered as feasible and accepted by people with ASD. This rationale is based on the clinical expertise of the development group members and available evidence on the effectiveness of sensory interventions in people with ASD (23, 24).	n.a.	n.a.	DG expert opinion
Peer support	People with ASD often experience challenges with sexual functions and intimate relationships. The development group considers peer support as effective to address these challenges in people with ASD. To the knowledge of the development group, no severe harms are associated with the use of peer support in ASD. The cost-benefit ratio in relation to the clinical benefits justifies the inclusion of this intervention in the Package of interventions for rehabilitation. Furthermore, peer	n.a.	n.a.	DG expert opinion

	support is considered as feasible and accepted by people with ASD. This rationale is based on the clinical expertise of the development group members.			
Movement functions				
Assessment of movement functions	Based on expert consensus, psychomotor and occupational therapy should be considered in case of comorbid developmental coordination disorder or other well specified motor problems that interfere with daily life, but only after clinical assessment and with regular re-assessments.	Conditional	Expert consensus	KCE (1): p. 103; Tab. 18:
Physical exercise training	Based on expert consensus, psychomotor and occupational therapy should be considered in case of comorbid developmental coordination disorder or other well specified motor problems that interfere with daily life, but only after clinical assessment and with regular re-assessments.	Conditional	Expert consensus	KCE (1): p. 103; Tab. 18:
Exercise tolerance functions				
Assessment of exercise capacity	Exercise capacity is often limited in people with ASD. The development group considers the assessment of exercise capacity as essential for the determination of intervention needs. To the knowledge of the development group, no severe harms are associated with the assessment. The cost-benefit ratio in relation to the clinical benefits justifies the inclusion of the assessment in the Package of interventions for rehabilitation. Furthermore, the assessment is considered as feasible and accepted by people with ASD.	n.a.	n.a.	DG expert opinion
Fitness training	Exercise capacity is often limited in people with ASD. The development group considers fitness training as effective to improve exercise capacity in people with ASD. To the knowledge of the development group, no severe harms are associated with performing fitness training in people with ASD. The cost-benefit ratio in relation to the clinical benefits justifies the inclusion of this intervention in the Package of interventions for rehabilitation. Furthermore, fitness training is considered as feasible and accepted by people with ASD. This rationale is based on the clinical expertise of the development group members and available evidence on the effectiveness of sensory interventions in people with ASD (25, 26).	n.a.	n.a.	DG expert opinion
Activities of daily living (ADL)				
Assessment of activities of daily living	Assess factors that may increase the risk of behaviour that challenges in routine assessment and care planning in children and young people with autism, including: - impairments in communication that may result in difficulty understanding situations or in expressing needs and wishes - coexisting physical disorders, such as pain or gastrointestinal disorders - coexisting mental health problems such as anxiety or depression and other neurodevelopmental conditions such as attention deficit hyperactivity disorder - the physical environment, such as lighting and noise levels - the social environment, including home, school and leisure activities - changes to routines or personal circumstances - developmental change, including puberty - exploitation or abuse by others - inadvertent reinforcement of behaviour that challenges - the absence of predictability and structure. When assessing challenging behaviour carry out a functional analysis (see recommendation 5.4.7.21) including identifying and evaluating any factors that may trigger or maintain the behaviour, such as: ● physical disorders ● the social environment (including relationships with family members, partners, carers and friends) ● the physical environment, including sensory factors ● coexisting mental disorders (including depression, anxiety disorders and psychosis) ● communication problems ● changes to routines or personal circumstances.	Strong	n.a.	NICE2 (3): p. 438; 7.7.1
		Strong	n.a.	NICE1 (2): p. 141; 5.4.7
ADL training	For adults with autism of all ranges of intellectual ability, who need help with activities of daily living (ADL), consider a structured and predictable training programme based on behavioural principles.	Conditional	Very Low	NICE1 (2): p. 207; 7.5.7
Provision and training in the use of assistive products for self-care	People with ASD often have limitations in carrying out ADLs independently. The development group considers the provision of assistive products for self-care as effective in addressing these limitations and facilitating the performance of ADLs in people with ASD. To the knowledge of the development group, no severe harms are associated with the use of assistive products for self-care in people with ASD. The cost-benefit ratio in relation to the clinical benefits justifies the	n.a.	n.a.	DG expert opinion

	inclusion of this intervention in the Package of interventions for rehabilitation. Furthermore, the provision and training of assistive products for self-care is considered as feasible and accepted by people with ASD.			
Modification of the home environment	Assess of home environment.	n.a.	n.a.	mhGAP (10): CHILD
Interpersonal interactions and relationships				
Assessment of interpersonal interactions and relationships	The assessment should make best use of existing documentation about personal, educational, occupational, social and communication functioning, and should include assessment of any coexisting conditions, especially depression, anxiety, attention deficit hyperactivity disorder, obsessive-compulsive disorder and global delay or intellectual disability in line with Autism in adults (NICE clinical guideline 142).	Strong	n.a.	NICE2 (3): p. 179; 5.5.1
	Healthcare professionals involved in specialist assessment should take an ASD-specific developmental history and should directly observe and assess the individual's social and communication skills and behaviour.	Strong	2+, 4	SIGN (4): p. 15; 4.2.3
Social skills training	For adults with autism without a learning disability or with a mild to moderate learning disability, who have identified problems with social interaction, consider: • a group-based social learning programme focused on improving social interaction • an individually delivered social learning programme for people who find group-based activities difficult.	Conditional	Low to very low	NICE1 (2): p. 231; 7.8.7
	Consider a specific social-communication intervention for the core features of autism in children and young people that includes play-based strategies with parents, carers, and teachers to increase joint attention, engagement and reciprocal communication in the child or young person. Strategies should: • be adjusted to the child or young person's developmental level • aim to increase the parents', carers', teachers', or peers' understanding of, and sensitivity and responsiveness to, the child or young person's patterns of communication and interaction • include techniques of therapist modelling and video-interaction feedback • include techniques to expand the child or young person's communication, interactive play and social routines. The intervention should be delivered by a trained professional. For preschool children consider parent, carer or teacher mediation. For school-aged children consider peer mediation.	Conditional	Low	NICE2 (3): p. 342; 6.6.1
Education				
Educational assessment	The assessment should make best use of existing documentation about personal, educational, occupational, social and communication functioning, and should include assessment of any coexisting conditions, especially depression, anxiety, attention deficit hyperactivity disorder, obsessive-compulsive disorder and global delay or intellectual disability in line with Autism in adults (NICE clinical guideline 142).	Strong	n.a.	NICE2 (3): p. 179; 5.5.1
Supported education	Access to and participation in education is often challenging for people with ASD. The development group considers supported education as effective to improve participation in education in people with ASD. To the knowledge of the development group, no severe harms are associated with the supported education in people with ASD. The cost-benefit ratio in relation to the clinical and social benefits justifies the inclusion of this intervention in the Package of interventions for rehabilitation. Furthermore, supported education is considered as feasible and accepted by people with ASD. This rationale is based on the clinical expertise of the development group members.	n.a.	n.a.	DG expert opinion
Provision and training in the use of assistive products for education	People with ASD often have limitations in performing activities related to education. The development group considers the provision and training in the use of assistive products for education as effective in addressing these limitations and facilitating the performance of activities related to education in people with ASD. To the knowledge of the development group, no severe harms are associated with the use of assistive products for education in people with ASD. The cost-benefit ratio in relation to the clinical benefits justifies the inclusion of this intervention in the Package of interventions for rehabilitation. Furthermore, the provision and training of assistive products for education is considered as feasible and accepted by people with ASD.	n.a.	n.a.	DG expert opinion

Work and employment				
Vocational assessment	The assessment should make best use of existing documentation about personal, educational, occupational, social and communication functioning, and should include assessment of any coexisting conditions, especially depression, anxiety, attention deficit hyperactivity disorder, obsessive-compulsive disorder and global delay or intellectual disability in line with Autism in adults (NICE clinical guideline 142).	Strong	Not reported	NICE2 (3): p. 179; 5.5.1
Vocational counselling, training, and support	Offer children and young people with autism support in developing coping strategies and accessing community services, including developing skills to access public transport, employment, and leisure facilities.	Strong	n.a.	NICE2 (3): p. 178; 5.5.1
Supported employment	For adults with autism without a learning disability or with a mild learning disability, who are having difficulty obtaining or maintaining employment, consider an individual supported employment programme.	Conditional	Very low	NICE1 (2): p. 254; 7.9.8
Provision and training in the use of assistive products for work	People with ASD often have limitations in performing activities related to vocation. The development group considers provision and training in the use of assistive products for work as effective in addressing these limitations and facilitating the performance of activities related to education in people with ASD. To the knowledge of the development group, no severe harms are associated with the use of assistive products for work in people with ASD. The cost-benefit ratio in relation to the clinical benefits justifies the inclusion of this intervention in the Package of interventions for rehabilitation. Furthermore, the provision and training of assistive products for work is considered as feasible and accepted by people with ASD.	n.a.	n.a.	DG expert opinion
Participation in community and social life				
Assessment of participation in community and social life	People with ASD often experience restrictions in participation in community and social life. The development group considers the assessment of participation in community and social life as essential for the determination of intervention needs. To the knowledge of the development group, no severe harms are associated with the assessment. The cost-benefit ratio in relation to the clinical benefits justifies the inclusion of the assessment in the Package of interventions for rehabilitation. Furthermore, the assessment is considered as feasible and accepted by people with ASD.	n.a.	n.a.	DG expert opinion
Peer support	People with ASD often experience restrictions in participation in community and social life. The development group considers peer support group as effective to improve participation in community and social life in people with ASD. To the knowledge of the development group, no severe harms are associated with peer support group in people with ASD. The cost-benefit ratio in relation to the clinical benefits justifies the inclusion of this intervention in the Package of interventions for rehabilitation. Furthermore, peer support is considered as feasible and accepted by people with ASD. This rationale is based on the clinical expertise of the development group members and available evidence on the effectiveness of peer support groups in people with ASD (27).	n.a.	n.a.	DG expert opinion
Participation focused interventions	For adults with autism without a learning disability or with a mild to moderate learning disability, who are socially isolated or have restricted social contact, consider: ● a group-based structured leisure activity programme ● an individually delivered structured leisure activity programme for people who find group-based activities difficult.	Conditional	Moderate to low	NICE1 (2): p. 222; 7.7.7
	Offer children and young people with autism support in developing coping strategies and accessing community services, including developing skills to access public transport, employment and leisure facilities.	Strong	n.a.	NICE2 (3): p. 178; 5.5.1
Self-management				
Education, advice and support for self-management of the health condition	Based on expert consensus adapted sexual education should be proposed to adolescents with autism.	Strong	Expert consensus	KCE (1): p. 104; Tab. 18

Carer and family support				
Assessment of carer and family needs	Offer families (including siblings) and carers an assessment of their own needs, including whether they have: • personal, social and emotional support • practical support in their caring role, including short breaks and emergency plans • a plan for future care for the child or young person, including transition to adult services.	Strong	n.a.	NICE2 (3): p. 178; 5.5.1
	During a comprehensive assessment, assess the following risks: • self-harm (in particular in people with depression or a moderate or severe learning disability) • rapid escalation of problems • harm to others • self-neglect • breakdown of family or residential support • exploitation or abuse by others. Develop a risk management plan if needed.	Strong	n.a.	NICE1 (2): p. 138; 5.4.7
Carer and family training and support	Caregiver skills training should be provided for management of children and adolescents with developmental disorders, including intellectual disabilities and pervasive developmental disorders (including autism).	Strong	Low	mhGAP (10): CHILD
	Behavioural interventions for children and adolescents, and caregiver skills training, may be offered for the treatment of behavioural disorders. Behavioural interventions (behavioural and cognitive behavioural therapies, school-based therapies, and caregiver skills training).	Conditional	Low	mhGAP (10): CHILD
	When the needs of families and carers have been identified, discuss help available locally and, taking into account their preferences, offer information, advice, training and support, especially if they: • need help with the personal, social or emotional care of the child or young person, including age-related needs such as self-care, relationships or sexuality • are involved in the delivery of an intervention for the child or young person in collaboration with health and social care professionals.	Strong	Very low to low	NICE2 (3): p. 178; 5.5.1
	Education and skills interventions for parents of preschool children with ASD should be offered.	Strong	1+, 4	SIGN (4): p. 41; 10.2
	Pharmacological treatment should be explained to the parents in a comprehensible way, if needed with written information on the therapeutic plan.	Strong	Expert consensus	KCE (1): p. 101; Tab. 18:
	Offer information, advice, training and support to families, partners, and carers if they: • need help with the personal, social or emotional care of the family member, partner or friend, or • are involved in supporting the delivery of an intervention for their family member, partner or friend (in collaboration with professionals).	Strong	Very low	NICE1 (2): p. 260; 7.10.7
	Give all families, partners, and carer(s) (whether or not the person wants them to be involved in their care) verbal and written information about: • autism and its management • local support groups and services specifically for families, partners and carers • their right to a formal carer's assessment of their own physical and mental health needs, and how to access this.	Strong	Very low	NICE1 (2): p. 79; 4.3.11
	Offer all families (including siblings) and carers verbal and written information about their right to: • short breaks and other respite care • a formal carer's assessment of their own physical and mental health needs, and how to access these.	Strong	n.a.	NICE2 (3): p. 178; 5.5.1
	Education and skills interventions for parents of preschool children with ASD should be offered.	Strong	1+, 4	SIGN (4): p. 41; 10.2
	Provide advice on age-appropriate stimulation and parenting.	n.a.	n.a.	mhGAP (10): CHILD
	Provide information regarding educational services and educate carer on importance of keeping the child/adolescent in school as much as possible.	n.a.	n.a.	mhGAP (10): CHILD
	Offer families, partners, and carers of adults with autism an assessment of their own needs including: • personal, social, and emotional support • support in their caring role, including despite care and emergency plans • advice on and support in obtaining practical support • planning of future care for the person with autism.	Strong	Very low	NICE1 (2): p. 260; 7.10.7
	Offer information, advice, training and support to families, partners, and carers if they: • need help with the personal, social or emotional care of the family member, partner or friend, or • are involved in supporting the delivery of an intervention for their family member, partner or friend (in collaboration with professionals).	Strong	Very low	NICE1 (2): p. 260; 7.10.7

	When the needs of families and carers have been identified, discuss help available locally and, taking into account their preferences, offer information, advice, training and support, especially if they: • need help with the personal, social or emotional care of the child or young person, including age-related needs such as self-care, relationships or sexuality • are involved in the delivery of an intervention for the child or young person in collaboration with health and social care professionals. Provide carer support.	Strong	n.a.	NICE2 (3): p. 178; 5.5.1
		n.a.	n.a.	mhGAP (10): CHILD
Mental health				
Assessment of mental health (in particular, depression and anxiety, and self-harm)	Assess factors that may increase the risk of behaviour that challenges in routine assessment and care planning in children and young people with autism, including: - impairments in communication that may result in difficulty understanding situations or in expressing needs and wishes - coexisting physical disorders, such as pain or gastrointestinal disorders - coexisting mental health problems such as anxiety or depression and other neurodevelopmental conditions such as ADHD - the physical environment, such as lighting and noise levels - the social environment, including home, school and leisure activities - changes to routines or personal circumstances - developmental change, including puberty - exploitation or abuse by others - inadvertent reinforcement of behaviour that challenges - the absence of predictability and structure. When assessing challenging behaviour carry out a functional analysis (see recommendation 5.4.7.21) including identifying and evaluating any factors that may trigger or maintain the behaviour, such as: • physical disorders • the social environment (including relationships with family members, partners, carers and friends) • the physical environment, including sensory factors • coexisting mental disorders (including depression, anxiety disorders and psychosis) • communication problems • changes to routines or personal circumstances. The assessment should make best use of existing documentation about personal, educational, occupational, social and communication functioning, and should include assessment of any coexisting conditions, especially depression, anxiety, attention deficit hyperactivity disorder, obsessive-compulsive disorder and global delay or intellectual disability in line with Autism in adults (NICE clinical guideline 142). During a comprehensive assessment, assess the following risks: • self-harm (in particular in people with depression or a moderate or severe learning disability) • rapid escalation of problems • harm to others • self-neglect • breakdown of family or residential support • exploitation or abuse by others. Develop a risk management plan if needed. Conducting an assessment for mental, neurological, and substance use (MNS) disorders involves the following steps. First, the presenting complaint is explored, then a history is obtained including asking about past MNS issues, general health problems, family MNS history, and psychosocial history. Observe the person (Mental Status Exam), establish a differential diagnosis, and identify the MNS condition. As part of the assessment, conduct a physical examination and obtain basic laboratory tests as needed. The assessment is conducted with informed consent of the person.	Strong	n.a.	NICE2 (3): p. 438; 7.7.1
		Strong	n.a.	NICE1 (2): p. 141; 5.4.7
		Strong	n.a.	NICE2 (3): p. 179; 5.5.1
		Strong	n.a.	NICE1 (2): p. 138; 5.4.7
		n.a.	n.a.	mhGAP (10) ^d
Antidepressants	When psychosocial interventions prove ineffective, fluoxetine (but not other Selective Serotonin Reuptake Inhibitors or Tricyclic Antidepressants) may be offered in adolescents with moderate-severe depressive episode/disorder. The intervention should only be offered under supervision of a specialist.	Conditional	Very low	mhGAP (10) ^d
Psychological therapies (incl. cognitive behavioural therapy)	Psychological interventions, such as cognitive behavioural therapy (CBT), interpersonal psychotherapy for children and adolescents with emotional disorders, and caregiver skills training focused on their caregivers, may be offered for the treatment of emotional disorders. CBT may be considered, using a group format where available and appropriate, to treat anxiety in children and young people with ASD and who have average verbal and cognitive ability. It is recommended to consider a cognitive-behavioural intervention to treat anxiety in children with autism who have the required verbal and cognitive ability to engage in CBT.	Conditional	Low	mhGAP (10) ^d
		Conditional	1++, 2++, 2+	SIGN (4): page 25; 6.3.3
		Conditional	Expert consensus	KCE (1): p. 103; Tab. 18

	Consider the following for children and young people with autism and anxiety who have the verbal and cognitive ability to engage in a CBT intervention: • group CBT adjusted to the needs of children and young people with autism • individual CBT for children and young people who find group-based activities difficult.	Conditional	n.a.	NICE2 (3): p. 592; 8.7.11
	Psychological interventions, such as CBT, interpersonal psychotherapy for children and adolescents with emotional disorders, and caregiver skills training focused on their caregivers, may be offered for the treatment of emotional disorders.	Conditional	Low	mhGAP (10) ^d
Physical exercise training	Advice on physical activity should be encouraged as part of treatment for adults with depressive episode/disorder with inactive lifestyles. In moderate and severe depression, this intervention should be considered as adjunct to antidepressants or brief structured psychological treatments.	Conditional	Very low	mhGAP (10) ^d
Stress management training	In non-specialized health care settings, relaxation training should be considered as treatment for anxiety symptoms (in absence of depressive episode/disorder) - who are in distress or have some degree of impaired functioning.	Conditional	Very low	mhGAP (10) ^d
	Relaxation training may be considered as treatment of adults with depressive episode/disorder. In moderate and severe depression, this intervention should be considered as adjunct to antidepressants or structured brief psychological treatments.	Conditional	Very low	mhGAP (10) ^d

^a See Table D1.6 for explanation of the strength of recommendation categories. ^b See Table D1.7. for explanation of the quality of evidence categories. ^c See Table D1.5. for reference to the clinical practice guideline. ^d See mhGAP Evidence Resource Center for original recommendations.

ADL: activities of daily living; ASD: autism spectrum disorders; DG: development group; n.a.: not available; MNS: mental, neurological, and substance use; PECS: Picture Exchange Communication System.

Table D1.10. Evidence table for evidence from Cochrane systematic reviews for assessments and interventions included in the *Package of interventions for rehabilitation for Autism Spectrum Disorders*

Assessments and interventions in the PIR	Intervention	Control	Outcomes	No. studies	No. participants	Corresponding risk (95% CI)	Relative risk (95% CI)	Heterogeneity	Quality of the evidence	Reference
Problems with behaviours										
Behavioural interventions	Early intensive behavioural intervention	Treatment as usual	Adaptive behaviour	5	202	MD 9.58 (5.57 to 13.60)	-	$I^2 = 0.0\%$; $P = 0.66$	Low	Reichow et al. (12)
			Problem behaviour	2	67	MD -0.58 (-1.24 to 0.07)	-	$I^2 = 41\%$; $P = 0.19$	Very low	
Communication										
Provision and training in the use of assistive products for communication	Picture Exchange Communication System (PECS)	Control	Spoken communication	1	84	-	OR 1.10 (0.46 to 2.62)	-	Very low	Brignell et al. (11)
			Non-verbal communication			-	OR 3.90 (1.75 to 8.68)	-	Very low	
			Combined spoken and non-verbal communication			-	OR 2.73 (1.22 to 6.08)	-	Very low	
			Social communication and pragmatic language skills			-	OR 0.55 (0.25 to 1.19)	-	Very low	

Interpersonal interactions and relationships										
Social skills training	Social skills groups	No intervention, wait list, or treatment as usual.	Social competence	4	178	SMD 0.47 (0.16 to 0.78)	-	$I^2 = 5\%$; $P = 0.37$	Low	Reichow et al. (13)
			Social communication	1	34	SMD 0.05 (-0.63 to 0.72)	-	-	Low	
			Emotion recognition	2	54	SMD 0.34 (-0.20 to 0.88)	-	$I^2 = 0.0\%$; $P = 0.63$	Low	
			Friendship	2	101	SMD 0.41 (0.02 to 0.81)	-	$I^2 = 0.0\%$; $P = 0.33$	Low	
			Loneliness	1	68	SMD -0.66 (-1.15 to -0.17)	-	-	Low	

CI: confidence interval; MD: mean difference; No: number; OR: odds ratio; PIR: package of interventions for rehabilitation; SMD: standardized mean difference.

1.5. References

1. Management of autism in children and young people: a good clinical practice guideline. Brussels: Belgian Health Care Knowledge Centre. 2014 (<https://kce.fgov.be/en/management-of-autism-in-children-and-young-people-a-good-clinical-practice-guideline>, accessed February 2023).
2. Autism. Recognition, referral, diagnosis and management of adults on the autism spectrum (NICE CG142). London: National Institute for Health and Care Excellence (NICE); 2012 (<https://www.nice.org.uk/guidance/cg142>, accessed February 2023).
3. Autism. The management and support of children and young people on the autism spectrum (CG 170). London: National Institute for Health and Care Excellence (NICE); 2013 (<https://www.ncbi.nlm.nih.gov/books/NBK299062/>, accessed February 2023).
4. Assessment, diagnosis and interventions for autism spectrum disorders (SIGN 145). Edinburgh: Healthcare Improvement Scotland: Scottish Intercollegiate Guidelines Network (SIGN); 2016 (<https://www.sign.ac.uk/our-guidelines/assessment-diagnosis-and-interventions-for-autism-spectrum-disorders/>, accessed February 2022).
5. New Zealand autism spectrum disorder guideline. Wellington: Ministries of Health and Education. 2016 (<https://www.health.govt.nz/publication/new-zealand-autism-spectrum-disorder-guideline>, accessed February 2023).
6. Management of autism spectrum disorder in children and adolescents. Putrajaya: Malaysian Health Technology Assessment Section (MaHTAS). 2018; ([https://www.moh.gov.my/moh/resources/maklumat%20terkini/CPG Mx of ASD in children and adolescents draft.pdf](https://www.moh.gov.my/moh/resources/maklumat%20terkini/CPG_Mx_of_ASD_in_children_and_adolescents_draft.pdf), accessed February 2023).
7. Clinical practice guideline on assessment and intervention services for young children with autism spectrum disorders (ASD). New York: New York State Department of Health; 2017 Update. (<https://www.health.ny.gov/publications/20152.pdf>, accessed February 2023).
8. Findings and conclusions: National Standards Project; Phase 2. Addressing the need for evidence-based practice guidelines for autism spectrum disorder. Randolph, Massachusetts: National Autism Center; 2015 (<https://www.autismdiagnostics.com/assets/Resources/NSP2.pdf>, accessed February 2023).
9. Appraisal of Guidelines for Research & Evaluation. The AGREE Next Steps Consortium. The AGREE II Instrument. Update 2017. The Agree Research Trust 2009. (<https://www.agreetrust.org/wp-content/uploads/2017/12/AGREE-II-Users-Manual-and-23-item-Instrument-2009-Update-2017.pdf>, accessed 8 June 2023).
10. mhGAP Intervention guide for mental, neurological and substance use disorders in non-specialized health settings. Version 2.0, Geneva: World Health Organization; 2016 (<https://apps.who.int/iris/handle/10665/250239>, accessed October 2022).
11. Brignell A, Chenausky K, Song H, Zhu J, Suo C, Morgan A. Communication interventions for autism spectrum disorder in minimally verbal children. Cochrane Database Syst Rev. 2018;11(11):CD012324.
12. Reichow B, Hume K, Barton E, Boyd B. Early intensive behavioral intervention (EIBI) for young children with autism spectrum disorders (ASD). Cochrane Database Syst Rev. 2018;5(5):CD009260.
13. Reichow B, Steiner A, Volkmar F. Social skills groups for people aged 6 to 21 with autism spectrum disorders (ASD). Cochrane Database Syst Rev. 2012;(7):CD008511.
14. Teh E, Vijayakumar R, Tan T, Yap M. Effects of physical exercise interventions on stereotyped motor behaviours in children with ASD: a meta-analysis. J Autism Dev Disord. 2022;52(7):2934–57.
15. Ferreira J, Ghiarone T, Júnior C, Furtado G, Carvalho H, Rodrigues A, et al. Effects of physical exercise on the stereotyped behavior of children with autism spectrum disorders. Medicina (Kaunas). 2019;55(10):685.

16. Bremer E, Crozier M, Lloyd M. A systematic review of the behavioural outcomes following exercise interventions for children and youth with autism spectrum disorder. *Autism*. 2016;20(8):899–915.
17. Schoen S, Lane S, Mailloux Z, May-Benson T, Parham L, Smith Roley S, et al. A systematic review of ayres sensory integration intervention for children with autism. *Autism Res*. 2019;12(1):6–19.
18. Weitlauf A, Sathe N, McPheeters M, Warren Z. Interventions targeting sensory challenges in autism spectrum disorder: a systematic review. *Pediatrics*. 2017;139(6):e20170347.
19. Case-Smith J, Weaver L, Fristad M. A systematic review of sensory processing interventions for children with autism spectrum disorders. *Autism*. 2015;19(2):133–48.
20. Sharp W, Berry R, McCracken C, Nuhu N, Marvel E, Saulnier C, et al. Feeding problems and nutrient intake in children with autism spectrum disorders: a meta-analysis and comprehensive review of the literature. *J Autism Dev Disord*. 2013;43(9):2159–73.
21. Marshall J, Ware R, Ziviani J, Hill RJ, Dodrill P. Efficacy of interventions to improve feeding difficulties in children with autism spectrum disorders: a systematic review and meta-analysis. *Child Care Health Dev*. 2015;41(2):278–302.
22. Marí-Bauset S, Zazpe I, Mari-Sanchis A, Llopis-González A, Morales-Suárez-Varela M. Food selectivity in autism spectrum disorders. *J Child Neurol*. 2014;29(11):1554–61.
23. Hancock G, Stokes M, Mesibov G. Socio-sexual functioning in autism spectrum disorder: a systematic review and meta-analyses of existing literature. *Autism Res*. 2017;10(11):1823–33.
24. Pecora L, Hooley M, Sperry L, Mesibov G, Stokes M. Sexuality and gender issues in individuals with autism spectrum disorder. *Child Adolesc Psychiatr Clin N Am*. 2020;29(3):543–56.
25. Fang Q, Aiken CA, Fang C, Pan Z. Effects of exergaming on physical and cognitive functions in individuals with autism spectrum disorder: a systematic review. *Games Health J*. 2019;8(2):74–84.
26. Srinivasan S, Pescatello L, Bhat A. Current perspectives on physical activity and exercise recommendations for children and adolescents with autism spectrum disorders. *Phys Ther*. 2014;94(6):875–89.
27. Chang Y, Locke J. A systematic review of peer-mediated interventions for children with autism spectrum disorder. *Res Autism Spectr Disord*. 2016; 27:1–10.

2. Package of interventions for rehabilitation for disorders of intellectual development

2.1. Literature search for clinical practice guidelines for rehabilitation in disorders of intellectual development

Table D2.1. Searched academic databases and search terms

Academic databases	Search terms
PubMed	(guideline (tiab) OR guidelines(tiab) OR guideline (publication type) OR "Guidelines as Topic"(Mesh)) AND (intellectual disability(tiab) OR intellectual disability(mesh) OR mental retardation(mesh) OR intellectual disabilities(tiab) OR mental retardation(tiab) OR intellectual development disorder(tiab) OR intellectual development disorders(tiab) OR mental deficiency(tiab) OR mental deficiencies(tiab)) NOT ("Diagnosis"(Mesh) OR diagnosis(tiab) OR diagnostics(tiab) OR diagnosed(tiab))
EMBASE	('mental deficiency'/de OR 'mental deficiency':ab,ti OR 'mental deficiencies':ab,ti OR 'intellectual disability':ab,ti OR 'intellectual disabilities':ab,ti OR 'mental retardation':ab,ti OR 'intellectual development disorder':ab,ti OR 'intellectual development disorders':ab,ti) AND ('practice guideline':ab,ti OR guideline:ab,ti OR guidelines:ab,ti) NOT ('diagnosis'/exp OR 'diagnosis' OR diagnosis:ab,ti OR diagnostics:ab,ti OR diagnosed:ab,ti)
CINAHL PLUS	((MH "Practice Guidelines") OR TI (guideline OR guidelines) OR AB (guideline OR guidelines)) AND ((MH "Intellectual Disability") OR TI ("intellectual disability" OR "mental retardation" OR "intellectual disabilities" OR "intellectual development disorder" OR "intellectual development disorders" OR "mental deficiency" OR "mental deficiencies")) OR AB ("intellectual disability" OR "mental retardation" OR "intellectual disabilities" OR "intellectual development disorder" OR "intellectual development disorders" OR "mental deficiency" OR "mental deficiencies")) NOT ((MH "Diagnosis") OR TI (diagnosis OR diagnostics OR diagnosed) OR AB (diagnosis OR diagnostics OR diagnosed))
PEDro	Abstract & Title: "intellectual disability" Abstract & Title: "intellectual disabilities" Abstract & Title: "mental retardation" Abstract & Title: "mental deficiency" Abstract & Title: "mental deficiencies" Abstract & Title: "intellectual impairments" Abstract & Title: "intellectual impairment" Abstract & Title: "intellectual disability" OR "intellectual disabilities" Abstract & Title: "Intellectual Disability" "intellectual disabilities" "mental retardation" "intellectual development disorder" "intellectual development disorders" "mental deficiency" "mental deficiencies" "intellectual dysfunction" "intellectual impairment" "intellectual impairments" "mentally retarded" "mental disability" "mental disabilities" "mental handicap" "mental incapacity" "mentally handicapped" "mental retard" "mentally retarded" – When searched using OR
Google scholar	–

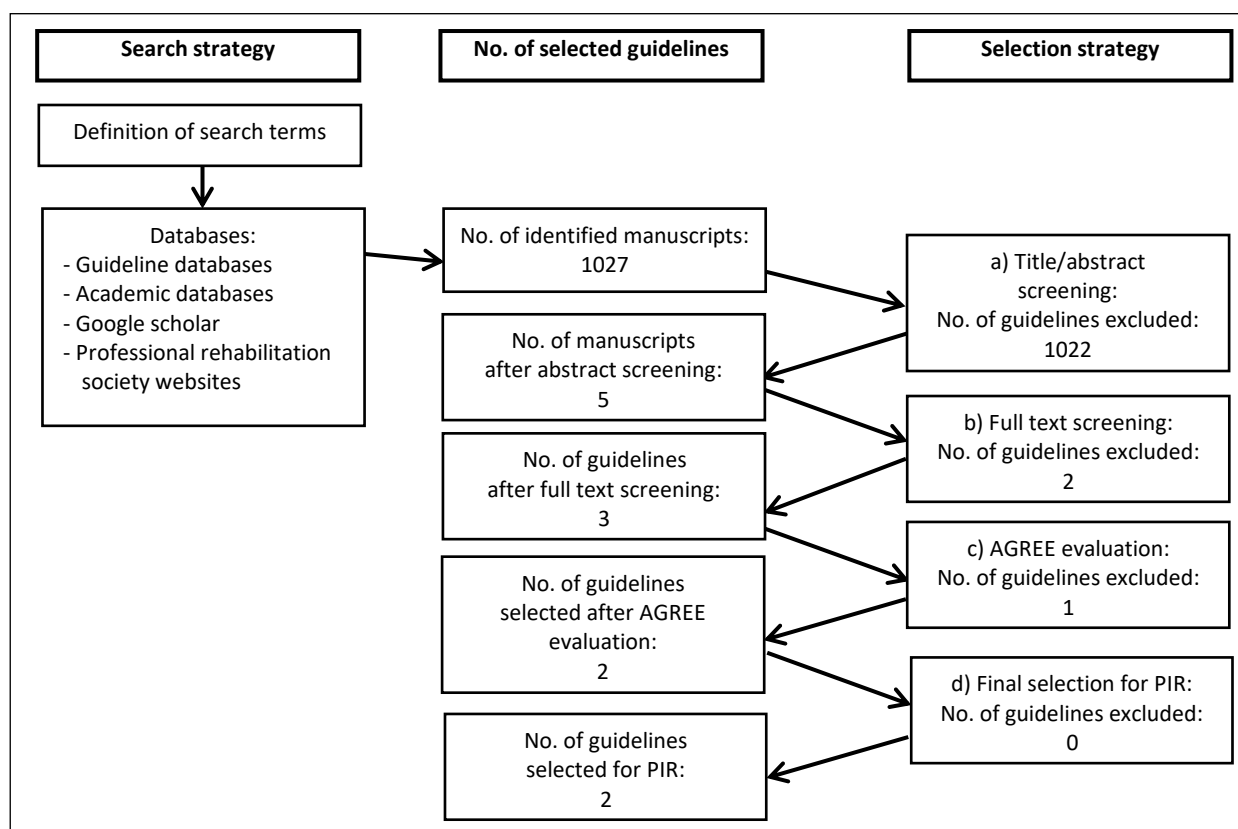
Table A2.2. Searched non-academic databases

Guideline databases
• Guidelines International Network (GIN) • National Institute for Health and Care Excellence (NICE, United Kingdom) • National Health and Medical Research Council (NHMRC, Australia) • Scottish Intercollegiate Guidelines Network (SIGN) • Canadian Medical Association (CMA) Infobase: Clinical Practice Guidelines • New Zealand Guidelines Group (NZGG) • National Guideline Clearinghouse (United States) • eGuidelines

Professional rehabilitation society websites

- American Academy of Neurology
- American Academy of Pediatrics
- American Academy of Child & Adolescent Psychiatry
- American Psychiatric Association
- American Occupational Therapy Association
- Society for Developmental & Behavioral Pediatrics
- American Speech-Language-Hearing Association

Fig. D2.1 Flow chart for the search and selection of clinical practice guidelines for rehabilitation for disorders of intellectual development



AGREE: Appraisal of Guidelines for Research and Evaluation; No: number; PIR: package of interventions for rehabilitation.

2.2. Selection of clinical practice guidelines for disorders of intellectual development

Table D2.3. Results of the AGREE^a evaluation

Guidelines included for AGREE evaluation	Average of key items ^b				Average of summary value of items 4, 7, 8, 10, 12, 13, 15, 22, 23 ^b	Selection criteria fulfilled
	7	8	12	22		
National Institute for Health and Care Excellence (NICE): Mental health problems in people with learning disabilities: prevention, assessment, and management. NICE Guideline 54. 2016 (1).	7	6	7	5	54	Yes
National Institute for Health and Care Excellence (NICE): Challenging behaviour and learning disabilities: Prevention and interventions for people with learning disabilities whose behaviour challenges. NICE Guideline 11. 2015 (2).	7	7	7	5	54	Yes

Guidelines included for AGREE evaluation	Average of key items ^b				Average of summary value of items 4, 7, 8, 10, 12, 13, 15, 22, 23 ^b	Selection criteria fulfilled
	7	8	12	22		
Canadian Family Physician: Primary care of adults with intellectual and developmental disabilities: 2018 Canadian consensus guidelines (3).	2	2	4	6	31.33	No

^a AGREE (4).

^b Numbers highlighted in **bold** in the table indicate a criterion that has not been fulfilled.

Table D2.4. Final selection of clinical practice guidelines

Guidelines included after AGREE evaluation	Selection criteria			Finally selected
	Publication date	Comprehensive	Multi-professional	
National Institute for Health and Care Excellence (NICE): Mental health problems in people with learning disabilities: prevention, assessment and management. NICE Guideline 54. 2016 (1).	2016	Yes	Yes	Yes
National Institute for Health and Care Excellence (NICE): Challenging behavior and learning disabilities: Prevention and interventions for people with learning disabilities whose behavior challenges. NICE Guideline 11. 2015 (2).	2015	Yes	Yes	Yes

Additional WHO guideline on mental health interventions

mhGAP Intervention guide for mental, neurological and substance use disorders in non-specialized health settings: mental health GAP Action Programme (mhGAP) – version 2.0. 2016 (5).

Table D2.5. Information on the included clinical practice guidelines

Guideline included	Abbreviation ^a	Target population	Topic of the guideline
National Institute for Health and Care Excellence (NICE); Mental health problems in people with learning disabilities: prevention, assessment and management. 2016 (1).	MHID	Children, young people, and adults with mild, moderate, severe or profound learning disabilities and mental health problems, and their families and carers. People with genetic conditions associated with learning disabilities and mental health problems, if some of their mental health problems and needs may differ from those of people with other learning disabilities (for example, Down's syndrome, Prader-Willi syndrome, Fragile X syndrome).	Prevention, identification, assessment and management of mental health problems in people with learning disabilities.
National Institute for Health and Care Excellence (NICE): Challenging Behaviour and Learning Disabilities: Prevention and interventions for people with learning disabilities whose behaviour challenges. 2015 (2).	CBID	Children, young people, and adults with learning disabilities and challenging behaviours.	General principles of care, support and interventions for family members or carers, early identification of the emergence of behaviour that challenges, assessment, psychological and environmental interventions, medication, interventions for coexisting health problems and sleep problems

Guideline included	Abbreviation*	Target population	Topic of the guideline
WHO Update of the Mental Health Gap Action Programme (mhGAP) Guideline for Mental, Neurological and Substance use Disorders. 2015 (5).	mhGAP	People with mental, neurological, and substance use disorders	mhGAP-CHILD: mental health interventions for children and adolescents with mental and behavioural disorders mhGAP-DEP: mental health interventions for depression mhGAP-OTH: mental health interventions for other significant emotional and medical unexplained somatic complaints.

^a Abbreviated titles are used in the evidence tables as references to the corresponding guidelines.

Table D2.6. Classification of the strength of recommendation in the included clinical practice guideline

Clinical practice guideline	Strength of recommendation		
	Strong	Weak/Conditional	Expert opinion
MHID (1) CBID (2)	Strong in favour: Recommendations on interventions that 'should' be used, the guideline development group is confident that, for the vast majority of people, the intervention (or interventions) will do more good than harm.	Conditional: Recommendations on interventions that should be 'considered', the guideline development group is confident that the intervention will do more good than harm for most patients. The choice of intervention is therefore more likely to vary depending on a person's values and preferences, and so the healthcare professional should spend more time discussing the options with the patient.	-
mhGAP (5)	Strong: Recommendation suggests that the GDG agreed that the quality of the evidence combined with certainty about the values and preferences and the feasibility of the recommendation meant it should be followed in all or almost all circumstances.	Conditional: Recommendation suggests less certainty about the quality of evidence and variation values and preferences and feasibility, leading to circumstances in which the recommendation may not apply.	Good practice points: Recommended best practice based on the clinical experience of the guideline development group.

CBID: Challenging behaviour and learning disabilities (2); mhGAP: WHO Mental Health Gap Action Programme Intervention Guide (5); MHID: Mental health problems in people with learning disabilities (1).

Table D2.7. Classification of the quality of the evidence in the included clinical practice guidelines

Clinical practice guideline	Quality of the evidence				
	High	Moderate	Low	Very low	Not classified/not reported
MHID (1) CBID (2)	High: High confidence in the correlation between true and estimated effect.	Moderate: Moderate confidence in the estimated effect. It is possible that the true effect is very different from the estimated effect.	Low: Limited confidence in the estimated effect. The true effect may be very different from the estimated effect.	Very low: Very little confidence in the estimated effect. The true effect is very probably different from the estimated effect.	Limited evidence: The development group did not assign a rating for the quality of the evidence. The group only stated that the evidence was limited. Not applicable: The development group did not assign a rating for the quality of the evidence because no clinical evidence was found that met the inclusion criteria.
mhGAP (5)	High: High confidence in the correlation between true and estimated effect.	Moderate: Moderate confidence in the estimated effect. It is possible that the true effect is very different from the estimated effect.	Low: Limited confidence in the estimated effect. The true effect may be very different from the estimated effect.	Very low: Very little confidence in the estimated effect. The true effect is very probably different from the estimated effect.	Not reported: For some interventions, the quality of the evidence was not reported.

CBID: Challenging behaviour and learning disabilities (2); mhGAP: WHO Mental Health Gap Action Programme Intervention Guide (5); MHID: Mental health problems in people with learning disabilities (1).

2.3. Cochrane systematic reviews included for disorders of intellectual development

Table D2.8. Cochrane systematic reviews included for disorders of intellectual development

Author, title
Ali A, Hall I, Blickwedel J, Hassiotis A. Behavioural and cognitive-behavioural interventions for outwardly directed aggressive behaviour in people with intellectual disabilities. 2015. (6)

2.4. Evidence tables for disorders of intellectual development

Table D2.9. Evidence table for assessments and interventions included in the *Package of interventions for rehabilitation for disorders of intellectual development*

Assessments and intervention	Original recommendation from clinical practice guidelines /Development group members' rationales	Strength of recommendation ^a	Quality of the evidence ^b	Reference ^c
Cognitive functions				
Assessment of cognitive functions	Consider supplementing an assessment of dementia with an adult with learning disabilities with: · measures of symptoms, such as the Dementia Questionnaire for People with Learning Disabilities (DLD), the Down Syndrome Dementia Scale (DSDS) or the Dementia Screening Questionnaire for Individuals with Intellectual Disabilities (DSQIID) · measures of cognitive function to monitor changes over time, such as the Test for Severe Impairment (TSI) · measures of adaptive function to monitor changes over time.	Conditional	Limited evidence	MHID (1): p. 104; 4.6
Cognitive training	People with disorders of intellectual development (DID) often experience problems with cognitive functions. The development group considers cognitive training as an effective intervention to improve cognitive functions in people with DID. To the knowledge of the development group, no severe harms are associated with the use of cognitive training in DID. The cost-benefit ratio in relation to the clinical benefits justifies the inclusion of this intervention in the Package of interventions for rehabilitation. Furthermore, cognitive training is considered as feasible and accepted by people with DID. This rationale is based on the clinical expertise of the development group members and available evidence on the effectiveness of cognitive training in people with DID (7).	n.a.	n.a.	DG expert opinion
Provision and training in the use of assistive products for cognition	People with DID often experience problems with cognitive functions. Therefore, the development group considers the provision and training in the use of assistive products for cognitive functions as essential for people with DID. To the knowledge of the development group, no severe harms are associated with the provision of these assistive products. The cost-benefit ratio in relation to the clinical benefits justifies the inclusion of this intervention in the Package of interventions for rehabilitation. Furthermore, the use of assistive products for cognitive functions is considered as feasible and accepted by people with DID. This rationale is based on the clinical expertise of the development group members.	n.a.	n.a.	DG expert opinion
Sleep functions				
Assessment of sleep disturbances	Consider behavioural interventions for sleep problems in children, young people and adults with a learning disability and behaviour that challenges that consist of: · a functional analysis of the problem sleep behaviour to inform the intervention (for example, not reinforcing non-sleep behaviours) · structured bedtime routines.	Conditional	Very low to low	CBID (2): p. 256; 11.3.1.1

Behavioural interventions	Consider behavioural interventions for sleep problems in children, young people and adults with a learning disability and behaviour that challenges that consist of: · a functional analysis of the problem sleep behaviour to inform the intervention (for example, not reinforcing non-sleep behaviours) · structured bedtime routines.	Conditional	Very low to low	CBID (2): p. 256; 11.3.1.1
Problems with behaviour				
Assessment of problems with behaviour (incl. adaptive and challenging behaviour)	Assess recent changes in behaviour using information from family members, carers, staff or others involved in the assessment as well as information from relevant records and previous assessments. Take into account the nature, quality and length of their relationship with the person.	Strong	Limited evidence	MHID (1): p. 103; 4.6
	When conducting an assessment with a child or young person with learning disabilities, consider using tools such as the Developmental Behaviour Checklist – parent version (DBC-P) or the Strengths and Difficulties Questionnaire (SDQ).	Conditional	Limited evidence	MHID (1): p. 104; 4.6
	Complete a baseline assessment of adaptive behaviour with all adults with Down's syndrome.	Strong	Limited evidence	MHID (1): p. 104; 4.6
	Consider supplementing an assessment of dementia with an adult with learning disabilities with: · measures of symptoms, such as the Dementia Questionnaire for People with Learning Disabilities (DLD), the Down Syndrome Dementia Scale (DSDS) or the Dementia Screening Questionnaire for Individuals with Intellectual Disabilities (DSQIID) · measures of cognitive function to monitor changes over time, such as the Test for Severe Impairment (TSI) · measures of adaptive function to monitor changes over time.	Conditional	Limited evidence	MHID (1): p. 104; 4.6
	Consider using direct observation and recording or formal rating scales (for example, the Adaptive Behaviour Scale or Aberrant Behaviour Checklist) to monitor the development of behaviour that challenges.	Conditional	Very low to low	CBID (2): p. 151; 7.4
	If behaviour that challenges is emerging or apparent, or a family member, carer or member of staff (such as a teacher or care worker), has concerns about behaviour, carry out initial assessment that includes: · a description of the behaviour (including its severity, frequency, duration and impact on the person and others) from the person (if possible) and a family member, carer or a member of staff (such as a teacher or care worker) · an explanation of the personal and environmental factors involved in developing or maintaining the behaviour from the person (if possible) and a family member, carer or a member of staff (such as a teacher or care worker) · the role of the service, staff, family members or carers in developing or maintaining the behaviour. Consider using a formal rating scale (for example, the Aberrant Behaviour Checklist or Adaptive Behaviour Scale) to provide baseline levels for the behaviour and a scale (such as the Functional Analysis Screening Tool) to help understand its function.	Strong	Limited evidence	CBID (2): p. 167; 8.5.2
	As part of initial assessment of behaviour that challenges, take into account: · the person's abilities and needs (in particular, their expressive and receptive communication) · any physical or mental health problems, and the effect of medication, including side effects · developmental history, including neurodevelopmental problems (including the severity of the learning disability and the presence of autism or other behavioural phenotypes) · response to any previous interventions for behaviour that challenges · the impact of the behaviour that challenges on the person's: o quality of life and that of their family members or carers o independent living skills and educational or occupational abilities · social and interpersonal history, including relationships with family members, carers, staff (such as teachers) or other people with a learning disability (such as those the person lives with) · aspects of the person's culture that could be relevant to the behaviour that challenges · life history, including any history of trauma or abuse · recent life events and changes to routine · the person's sensory profile, preferences and needs · the physical environment, including heat, light, noise and smell · the care environment, including the range of activities available, how it engages people and promotes choice, and how well structured it is.	Strong	Limited evidence	CBID (2): p. 167; 8.5.
	Carry out a functional assessment of the behaviour that challenges to help inform decisions about interventions. This should include: · a clear description of the behaviour, including classes or sequences of behaviours that typically occur together · identifying the events, times and situations that predict when the behaviour will and will not occur across the full range of the person's daily routines and usual environments · identifying the consequences (or reinforcers) that	Strong	Limited evidence	CBID (2): p. 170; 8.5.5

	maintain the behaviour (that is, the function or purpose that the behaviour serves) · developing summary statements or hypotheses that describe the relationships between personal and environmental triggers, the behaviour and its reinforcers · collecting direct observational data to inform the summary statements or hypotheses.			
	Include the following in a functional assessment: · a baseline measurement of current behaviour, and its frequency and intensity, and repeated measurements in order to evaluate change · measurements including direct observations and scales such as the Aberrant Behaviour Checklist and self-reporting · a baseline measurement of quality of life (such as the Life Experiences Checklist and the Quality of Life Questionnaire) · assessment of the impact of current or past interventions, including reactive strategies.	Strong	Limited evidence	CBID (2): p. 170; 8.5.5
	Vary the complexity and intensity of the functional assessment according to the complexity and intensity of behaviour that challenges, following a phased approach as set out below. · Carry out pre-assessment data gathering to help shape the focus and level of the assessment. · For recent-onset behaviour that challenges, consider brief structured assessments such as the Functional Analysis Screening Tool or Motivation Assessment Scale to identify relationships between the behaviour and what triggers and reinforces it. · For recent-onset behaviour that challenges, or marked changes in patterns of existing behaviours, take into account whether any significant alterations to the person's environment and physical or psychological health are associated with the development or maintenance of the behaviour. · Consider in-depth assessment involving interviews with family members, carers and others, direct observations, structured record keeping, questionnaires and reviews of case records. · If a mental health problem may underlie behaviour that challenges, consider initial screening using assessment scales such as the Diagnostic Assessment Schedule for the Severely Handicapped-II, Psychiatric Assessment Schedule for Adults with a Developmental Disability or the Psychopathology Instrument for Mentally Retarded Adults and seek expert opinion.	Conditional	Limited evidence	CBID (2): p. 170; 8.5.5
Cognitive behavioural therapy	Behavioural interventions for children and adolescents, and caregiver skills training, may be offered for the treatment of behavioural disorders. Behavioural interventions (behavioural and cognitive behavioural therapies, school-based therapies, and caregiver skills training).	Conditional	Low	mhGAP (5): CHILD
	Consider individual psychological interventions for adults with an anger management problem. These interventions should be based on cognitive-behavioural principles and delivered individually or in groups over 15–20 hours.	Conditional	Very low to low	CBID (2): p. 255; 11.3.1
Behavioural interventions	Behavioural interventions for children and adolescents, and caregiver skills training, may be offered for the treatment of behavioural disorders. Behavioural interventions (behavioural and cognitive behavioural therapies, school-based therapies, and caregiver skills training).	Conditional	Low	mhGAP (5): CHILD
	Develop a written behaviour support plan for children, young people and adults with a learning disability and behaviour that challenges that is based on a shared understanding about the function of the behaviour. This should: · identify proactive strategies designed to improve the person's quality of life and remove the conditions likely to promote behaviour that challenges, including: o changing the environment (for example, reducing noise, increasing predictability) o promoting active engagement through structured and personalized daily activities, including adjusting the school curriculum for children and young people · identify adaptations to a person's environment and routine, and strategies to help them develop an alternative behaviour to achieve the function of the behaviour that challenges by developing a new skill (for example, improved communication, emotional regulation or social interaction) · identify preventive strategies to calm the person when they begin to show early signs of distress, including: o individual relaxation techniques o distraction and diversion onto activities they find enjoyable and rewarding · identify reactive strategies to manage any behaviours that are not preventable (see section 13.3), including how family members, carers or staff should respond if a person's agitation escalates and there is a significant risk of harm to them or others · incorporate risk management and take into account the effect of the behaviour support plan on the level of risk · be compatible with	Strong	Limited evidence	CBID (2): p. 171; 8.5.7

	the abilities and resources of the person's family members, carers or staff, including managing risk, and can be implemented within these resources · be supported by data that measure the accurate implementation of the plan · be monitored using the continuous collection of objective outcome data · be reviewed frequently (fortnightly for the first 2 months and monthly thereafter), particularly if behaviour that challenges or use of restrictive interventions increases, or quality of life decreases · identify any training for family members, carers or staff to improve their understanding of behaviour that challenges shown by people with a learning disability · identify those responsible for delivering the plan and the designated person responsible for coordinating it.			
Relaxation training	Develop a written behaviour support plan for children, young people and adults with a learning disability and behaviour that challenges that is based on a shared understanding about the function of the behaviour. This should: · identify proactive strategies designed to improve the person's quality of life and remove the conditions likely to promote behaviour that challenges, including: · identify preventive strategies to calm the person when they begin to show early signs of distress, including: · individual relaxation techniques · distraction and diversion onto activities they find enjoyable and rewarding · identify reactive strategies to manage any behaviours that are not preventable (see section 13.3), including how family members, carers or staff should respond if a person's agitation escalates and there is a significant risk of harm to them or others · incorporate risk management and take into account the effect of the behaviour support plan on the level of risk · be compatible with the abilities and resources of the person's family members, carers or staff, including managing risk, and can be implemented within these resources · be supported by data that measure the accurate implementation of the plan · be monitored using the continuous collection of objective outcome data · be reviewed frequently (fortnightly for the first 2 months and monthly thereafter), particularly if behaviour that challenges or use of restrictive interventions increases, or quality of life decreases · identify any training for family members, carers or staff to improve their understanding of behaviour that challenges shown by people with a learning disability · identify those responsible for delivering the plan and the designated person responsible for coordinating it.	Strong	Limited evidence	CBID (2): p. 171; 8.5.7
Perceptual functions				
Assessment of perceptual functions	Sensory processing dysfunctions often exist in people with DID. The development group considers the assessment of perceptual functions as essential for the determination of intervention needs. To the knowledge of the development group, no severe harms are associated with the assessment. The cost-benefit ratio in relation to the clinical benefits justifies the inclusion of the assessment in the Package of interventions for rehabilitation. Furthermore, the assessment is considered as feasible and accepted by people with DID. This rationale is based on the clinical expertise of the development group members and evidence on the presence of sensory processing dysfunctions in people with DID (8).	n.a.	n.a.	DG expert opinion
Sensory integration interventions	Sensory processing dysfunctions often exist in people with DID. The development group considers sensory integration interventions as essential for people with DID. To the knowledge of the development group, no severe harms are associated with the application of sensory integration interventions in DID. The cost-benefit ratio in relation to the clinical benefits justifies the inclusion of this intervention in the Package of interventions for rehabilitation. Furthermore, sensory integration interventions are considered as feasible and accepted by people with DID.	n.a.	n.a.	DG expert opinion
Provision and training in the use of assistive products for perceptual functions	People with DID often experience problems with sensory perception and processing. The development group considers the provision and training in the use of assistive products for sensory perception challenges as an effective intervention to address these limitations. To the knowledge of the development group, no severe harms are associated with the provision of assistive products. The cost-benefit ratio in relation to the clinical benefits justifies the inclusion of the assessment in the Package of interventions for rehabilitation. Furthermore, the provision of assistive products is considered as feasible and accepted by people with DID.	n.n.	n.a.	DG expert opinion
Cognitive functions of language				

Assessment of language	As part of initial assessment of behaviour that challenges, take into account: · the person's abilities and needs (in particular, their expressive and receptive communication) · any physical or mental health problems, and the effect of medication, including side effects · developmental history, including neurodevelopmental problems (including the severity of the learning disability and the presence of autism or other behavioural phenotypes) · response to any previous interventions for behaviour that challenges · the impact of the behaviour that challenges on the person's: o quality of life and that of their family members or carers o independent living skills and educational or occupational abilities · social and interpersonal history, including relationships with family members, carers, staff (such as teachers) or other people with a learning disability (such as those the person lives with) · aspects of the person's culture that could be relevant to the behaviour that challenges · life history, including any history of trauma or abuse · recent life events and changes to routine · the person's sensory profile, preferences and needs · the physical environment, including heat, light, noise and smell · the care environment, including the range of activities available, how it engages people and promotes choice, and how well structured it is.	Strong	Limited evidence	CBID (2): p. 167; 8.5.2
Language therapy	Cognitive functions of language are often limited in people with DID. Therefore, the development group considers language therapy as essential for people with DID. To the knowledge of the development group, no severe harms are associated with the use of speech and language therapy in people with DID. The cost-benefit ratio in relation to the clinical benefits justifies the inclusion of this intervention in the Package of interventions for rehabilitation. Furthermore, speech and language therapy is considered as feasible and accepted by people with intellectual disability.	n.a.	n.a.	DG expert opinion
Speech functions				
Assessment of speech functions	The production of voice and speech is often limited in people with DID. Therefore, the development group considers the assessment of speech functions as essential for the determination of intervention needs. To the knowledge of the development group, no severe harms are associated with the assessment. The cost-benefit ratio in relation to the clinical benefits justifies the inclusion of the assessment in the Package of interventions for rehabilitation. Furthermore, the assessment is considered as feasible and accepted by people with DID.	n.a.	n.a.	DG expert opinion
Speech therapy	The production of voice and speech is often limited in people with DID. Therefore, the development group considers speech therapy as essential for people with DID. To the knowledge of the development group, no severe harms are associated with the use of speech therapy in people with DID. The cost-benefit ratio in relation to the clinical benefits justifies the inclusion of this intervention in the Package of interventions for rehabilitation. Furthermore, speech therapy is considered as feasible and accepted by people with DID.	n.a.	n.a.	DG expert opinion
Communication				
Assessment of communication	Everyone involved in caring for and supporting children, young people and adults with a learning disability (including family members and carers) should understand the risk of behaviour that challenges and that it often develops gradually. Pay attention to and record factors that may increase this risk, including: · personal factors, such as o a severe learning disability o autism o dementia o communication difficulties (expressive and receptive) o visual impairment (which may lead to increased self-injury and stereotypy) o physical health problems o variations with age (peaking in the teens and twenties) · environmental factors, such as: o abusive or restrictive social environments o with little or too much sensory stimulation and those with low engagement levels (for example, little interaction with staff) o developmentally inappropriate environments (for example, a curriculum that makes too many demands on a child or young person) o environments where disrespectful social relationships and poor communication are typical or where staff do not have the capacity or resources to respond to people's needs o changes to the person's environment (for example, significant staff changes or moving to a new care setting). Take into account the person's communication needs and level of understanding throughout assessments, treatment and care for a mental health problem, and: · speak to the person directly rather than talking about or over them · use clear, straightforward and unambiguous language · assess whether communication aids, an advocate or someone	Strong	Very low to low	CBID (2): p. 150; 7.4
		Strong	Very low	MHID (1): p. 267; 8.2.2.

	familiar with the person's communication methods are needed · make adjustments to accommodate sensory impairments (including sight and hearing impairments) · explain the content and purpose of every meeting or session · use concrete examples, visual imagery, practical demonstrations and role play to explain concepts · communicate at a pace that is comfortable for the person, and arrange longer or additional meetings or treatment sessions if needed · use different methods and formats for communication (written, signing, visual, verbal, or a combination of these), depending on the person's preferences (see the Accessible Information Standard for guidance on ensuring people with learning disabilities receive information in formats they can understand) · regularly check the person's understanding · summarize and explain the conclusions of every meeting or session · check that the person has communicated what they wanted.			
Communication skills training	Preschool classroom-based interventions should have multiple components, including: · curriculum design and development · social and communication skills training for the children · skills training in behavioural strategies for parents or carers · training on how to mediate the intervention for preschool teachers.	Strong	Very low to low	CBID (2): p. 195; 9.4.1
Provision and training in the use of assistive products for communication	Take into account the person's communication needs and level of understanding throughout assessments, treatment and care for a mental health problem, and: · speak to the person directly rather than talking about or over them · use clear, straightforward and unambiguous language · assess whether communication aids, an advocate or someone familiar with the person's communication methods are needed · make adjustments to accommodate sensory impairments (including sight and hearing impairments) · explain the content and purpose of every meeting or session · use concrete examples, visual imagery, practical demonstrations and role play to explain concepts · communicate at a pace that is comfortable for the person, and arrange longer or additional meetings or treatment sessions if needed · use different methods and formats for communication (written, signing, visual, verbal, or a combination of these), depending on the person's preferences (see the Accessible Information Standard for guidance on ensuring people with learning disabilities receive information in formats they can understand) · regularly check the person's understanding · summarize and explain the conclusions of every meeting or session · check that the person has communicated what they wanted.	Strong	Very low	MHID (1): p. 267; 8.2.2.
Sexual functions and intimate relationships				
Assessment of sexual functions and intimate relationships	People with DID often experience challenges with sexual functions and intimate relationships. The development group considers the assessment of sexual functions and intimate relationships as essential for the determination of intervention needs. To the knowledge of the development group, no severe harms are associated with the assessment. The cost-benefit ratio in relation to the clinical benefits justifies the inclusion of the assessment in the Package of interventions for rehabilitation. Furthermore, the assessment is considered as feasible and accepted by people with DID.	n.a.	n.a.	DG expert opinion
Peer support	People with DID often experience challenges with sexual functions and intimate relationships. The development group considers peer support as effective to address these challenges in people with DID. To the knowledge of the development group, no severe harms are associated with the use of peer support in DID. The cost-benefit ratio in relation to the clinical benefits justifies the inclusion of this intervention in the Package of interventions for rehabilitation. Furthermore, peer support is considered as feasible and accepted by people with DID.	n.a.	n.a.	DG expert opinion
Movement functions				
Assessment of movement functions	People with DID often experience problems with movement functions. The development group considers the assessment of movement functions as essential for the determination of intervention needs. To the knowledge of the development group, no severe harms are associated with the assessment. The cost-benefit ratio in relation to the clinical benefits justifies the inclusion of the assessment in the Package of interventions for rehabilitation. Furthermore, the assessment is considered as feasible and accepted by people with DID. This rationale is based on the clinical	n.a.	n.a.	DG expert opinion

	expertise of the development group members and available evidence on the presence of movement disorders in people with DID (9).			
Physical exercise training	Movement functions are often impaired in people with DID. The development group considers physical exercise training as an essential intervention to improve movement functions in people with DID. To the knowledge of the development group, no severe harms are associated with the use of physical exercise training in DID. The cost-benefit ratio in relation to the clinical benefits justifies the inclusion of this intervention in the Package of interventions for rehabilitation. Furthermore, the assessment is considered as feasible and accepted by people with DID. This rationale is based on the clinical expertise of the development group members and available evidence on the effectiveness of physical exercise training in people with DID (10).	n.a.	n.a.	DG expert opinion
Exercise tolerance functions				
Assessment of exercise capacity	People with DID often experience problems with exercise capacity. The development group considers the assessment of exercise capacity as essential for the determination of intervention needs. To the knowledge of the development group, no severe harms are associated with the assessment. The cost-benefit ratio in relation to the clinical benefits justifies the inclusion of the assessment in the Package of interventions for rehabilitation. Furthermore, the assessment is considered as feasible and accepted by people with DID.	n.a.	n.a.	DG expert opinion
Fitness training	Exercise capacity is often limited in people with DID. The development group considers fitness training as essential for people with DID. To the knowledge of the development group, no severe harms are associated with performing fitness training in people with DID. The cost-benefit ratio in relation to the clinical benefits justifies the inclusion of this intervention in the Package of interventions for rehabilitation. Furthermore, fitness training is considered as feasible and accepted by people with DID. This rationale is based on the clinical expertise of the development group members and available evidence on fitness training in people with DID (11).	n.a.	n.a.	DG expert opinion
Activities of daily living (ADL)				
Assessment of activities of daily living	Consider supplementing an assessment of dementia with an adult with learning disabilities with: · measures of symptoms, such as the Dementia Questionnaire for People with Learning Disabilities (DLD), the Down Syndrome Dementia Scale (DSDS) or the Dementia Screening Questionnaire for Individuals with Intellectual Disabilities (DSQIID) · measures of cognitive function to monitor changes over time, such as the Test for Severe Impairment (TSI) · measures of adaptive function to monitor changes over time.	Conditional	Limited evidence	MHID (1): pg 104; 4.6
	Complete a baseline assessment of adaptive behaviour with all adults with Down's syndrome.	Strong	Limited evidence	MHID (1): pg 104; 4.6
ADL training	Performing ADL is often limited in people with DID. The development group considers ADL training as effective to improve ADL in people with DID. To the knowledge of the development group, no severe harms are associated with performing ADL training in people with DID. The cost-benefit ratio in relation to the clinical benefits justifies the inclusion of this intervention in the Package of interventions for rehabilitation. Furthermore, ADL training is considered as feasible and accepted by people with DID. This rationale is based on the clinical expertise of the development group members and available evidence on the effectiveness of ADL training in people with DID (12).	n.a.	n.a.	DG expert opinion
Provision and training in the use of assistive products for self-care	People with DID often have limitations in carrying out activities of daily living (ADL) independently. The development group considers the provision of assistive products for self-care as effective in addressing these limitations and facilitating the performance of ADL in people with DID. To the knowledge of the development group, no severe harms are associated with the use of assistive products for self-care in people with DID. The cost-benefit ratio in relation to the clinical benefits justifies the inclusion of this intervention in the Package of interventions for rehabilitation. Furthermore, the provision and training of assistive products for self-care is considered as feasible and accepted by people with DID.	n.a.	n.a.	DG expert opinion

Modification of the home environment	Performing activities of daily living is often limited in people with DID. The development group considers modification of the home environment as effective to improve activities of daily living in people with DID. To the knowledge of the development group, no severe harms are associated with modification of the home environment. The cost-benefit ratio in relation to the clinical benefits justifies the inclusion of this intervention in the Package of interventions for rehabilitation. Furthermore, environmental modifications are considered as feasible and accepted by people with DID.	n.a.	n.a.	DG expert opinion
Interpersonal interactions and relationships				
Assessment of interpersonal interactions and relationships	As part of initial assessment of behaviour that challenges, take into account: · the person's abilities and needs (in particular, their expressive and receptive communication) · any physical or mental health problems, and the effect of medication, including side effects · developmental history, including neurodevelopmental problems (including the severity of the learning disability and the presence of autism or other behavioural phenotypes) · response to any previous interventions for behaviour that challenges · the impact of the behaviour that challenges on the person's: o quality of life and that of their family members or carers o independent living skills and educational or occupational abilities · social and interpersonal history, including relationships with family members, carers, staff (such as teachers) or other people with a learning disability (such as those the person lives with) · aspects of the person's culture that could be relevant to the behaviour that challenges · life history, including any history of trauma or abuse · recent life events and changes to routine · the person's sensory profile, preferences and needs · the physical environment, including heat, light, noise and smell · the care environment, including the range of activities available, how it engages people and promotes choice, and how well structured it is.	Strong	Limited evidence	CBID (2): p. 167; 8.5.2
Social skills training	Preschool classroom-based interventions should have multiple components, including: · curriculum design and development · social and communication skills training for the children · skills training in behavioural strategies for parents or carers · training on how to mediate the intervention for preschool teachers.	Strong	Very low to low	CBID (2): p. 195; 9.4.1
Education				
Educational assessment	Assess the school environment.	n.a.	n.a.	mhGAP (5): CHILD
	Everyone involved in caring for and supporting children, young people and adults with a learning disability (including family members and carers) should understand the risk of behaviour that challenges and that it often develops gradually. Pay attention to and record factors that may increase this risk, including: · personal factors, such as o a severe learning disability · autism o dementia o communication difficulties (expressive and receptive) · visual impairment (which may lead to increased self-injury and stereotypy) · physical health problems · variations with age (peaking in the teens and twenties) · environmental factors, such as: · abusive or restrictive social environments · with little or too much sensory stimulation and those with low engagement levels (for example, little interaction with staff) · developmentally inappropriate environments (for example, a curriculum that makes too many demands on a child or young person) · environments where disrespectful social relationships and poor communication are typical or where staff do not have the capacity or resources to respond to people's needs · changes to the person's environment (for example, significant staff changes or moving to a new care setting).	Strong	Very low to low	CBID (2): p. 150; 7.4
Supported education	People with DID often experience restrictions in enrolment and participation in education. The development group considers supported education as an effective intervention to improve participation in education in people with DID. To the knowledge of the development group, no severe harms are associated with supported education in people with DID. The cost-benefit ratio in relation to the clinical benefits justifies the inclusion of this intervention in the Package of interventions for rehabilitation. Furthermore, supported education is considered as feasible and accepted by people with DID. This rationale is based on the clinical expertise of the development group members.	n.a.	n.a.	DG expert opinion

Provision and training in the use of assistive products for education	People with DID often have limitations in performing activities related to education. The development group considers the provision and training in the use of assistive products for education as effective in addressing these limitations and facilitating the performance of activities related to education in people with DID. To the knowledge of the development group, no severe harms are associated with the use of assistive products for education in people with DID. The cost-benefit ratio in relation to the clinical benefits justifies the inclusion of this intervention in the Package of interventions for rehabilitation. Furthermore, the provision and training of assistive products for education is considered as feasible and accepted by people with DID.	n.a.	n.a.	DG expert opinion
Work and employment				
Vocational assessment	As part of initial assessment of behaviour that challenges, take into account: · the person's abilities and needs (in particular, their expressive and receptive communication) · any physical or mental health problems, and the effect of medication, including side effects · developmental history, including neurodevelopmental problems (including the severity of the learning disability and the presence of autism or other behavioural phenotypes) · response to any previous interventions for behaviour that challenges · the impact of the behaviour that challenges on the person's: · and that of their family members or carers · independent living skills and educational or occupational abilities · social and interpersonal history, including relationships with family members, carers, staff (such as teachers) or other people with a learning disability (such as those the person lives with).	Strong	Limited evidence	CBID (2): p. 167; 8.5.2
Vocational counselling, training, and support	Actively encourage adults with learning disabilities (with or without a mental health problem) to find and participate in paid or voluntary work that is meaningful to them, if they are able.	Strong	n.a.	MHID (1): p. 248; 7.8.5
	Consider providing practical support to adults with learning disabilities (with or without a mental health problem) to find paid or voluntary work, including: · preparing a CV · identifying personal strengths and interests · completing application forms · preparing for interviews · accompanying the person to interviews · completing any pre-employment checks.	Conditional	n.a.	MHID (1): p. 248; 7.8.5
	Health and social care services should take account of an adult or young person's sensory, physical, cognitive and communication needs and the severity of their mental health problem (if any), and consider: · helping them to identify and overcome any possible challenges during employment · appointing supported employment workers to provide ongoing support to adults with learning disabilities and their employers · providing information and guidance to potential employers about the benefits of recruiting people with learning disabilities, · assisting employers in making reasonable adjustments to help them to work (in line with the Equality Act 2010).	Strong	n.a.	MHID (1): p. 248; 7.8.5
Supported employment	Health and social care services should take account of an adult or young person's sensory, physical, cognitive and communication needs and the severity of their mental health problem (if any), and consider: · helping them to identify and overcome any possible challenges during employment · appointing supported employment workers to provide ongoing support to adults with learning disabilities and their employers · providing information and guidance to potential employers about the benefits of recruiting people with learning disabilities, · assisting employers in making reasonable adjustments to help them to work (in line with the Equality Act 2010).	Strong	n.a.	MHID (1): p. 248; 7.8.5
Provision and training in the use of assistive products for work	People with DID often have limitations in performing activities related to vocation. The development group considers the provision and training in the use of assistive products for work as effective in addressing these limitations and facilitating the performance of activities related to education in people with DID. To the knowledge of the development group, no severe harms are associated with the use of assistive products for work in people with DID. The cost-benefit ratio in relation to the clinical benefits justifies the inclusion of this intervention in the Package of interventions for rehabilitation. Furthermore, the provision and training of assistive products for work is considered as feasible and accepted by people with DID.	n.a.	n.a.	DG expert opinion
Participation in community and social life				

Assessment of participation in community and social life	People with DID often experience restrictions in participation in community and social life. The development group considers the assessment of participation in community and social life as essential for the determination of intervention needs. To the knowledge of the development group, no severe harms are associated with the assessment. The cost-benefit ratio in relation to the clinical benefits justifies the inclusion of the assessment in the Package of interventions for rehabilitation. Furthermore, the assessment is considered as feasible and accepted by people with DID.	n.a.	n.a.	DG expert opinion
Peer support	In keeping with the preferences of the person with learning disabilities and mental health problems, all staff should support them to: · engage in community activities, such as going to a library or sports centre · access local community resources, such as libraries, cinemas, cafes and leisure centres · take part in leisure activities, such as hobbies, which are meaningful to the person. Reasonable adjustments may be needed to do this (in line with the Equality Act 2010), such as a buddy system, transport, or advising local facilities on accessibility.	Strong	n.a.	MHID (1): pg 248; 7.8.5
Participation focused interventions	In keeping with the preferences of the person with learning disabilities and mental health problems, all staff should support them to: · engage in community activities, such as going to a library or sports centre · access local community resources, such as libraries, cinemas, cafes and leisure centres · take part in leisure activities, such as hobbies, which are meaningful to the person. Reasonable adjustments may be needed to do this (in line with the Equality Act 2010), such as a buddy system, transport, or advising local facilities on accessibility.	Strong	n.a.	MHID (1): p. 248; 7.8.5
Self-management				
Assessment of decision making	Assess the person's capacity to make decisions throughout assessment, care and treatment for the mental health problem on a decision-by-decision basis, in accordance with the Mental Capacity Act and supporting codes of practice (see Your care). Help people make decisions by ensuring that their communication needs are met (see recommendation 61) and (if appropriate) involving a family member, carer, care worker or other individual familiar with the person's communication abilities.	Strong	n.a.	MHID (1): p. 266; 8.2.6
Education, advice and support for self-management of the health condition (incl. support in making decisions, living arrangements, prevent ongoing abuse)	Assess the person's capacity to make decisions throughout assessment, care and treatment for the mental health problem on a decision-by-decision basis, in accordance with the Mental Capacity Act and supporting codes of practice (see Your care). Help people make decisions by ensuring that their communication needs are met (see recommendation 61) and (if appropriate) involving a family member, carer, care worker or other individual familiar with the person's communication abilities.	Strong	n.a.	MHID (1): p. 266; 8.2.6
	Health, social care and education services should consider the impact of the social and physical environment on the mental health of adults with learning disabilities when developing care plans, and: · support people to live where and with whom they want · encourage family involvement in the person's life, if appropriate · support people to get involved in activities that are interesting and meaningful to them · plan for and help people with any significant changes to their living arrangements.	Strong	n.a.	MHID (1): p. 242; 7.7.4
	Provide information about where to seek help for any ongoing abuse.	n.a.	n.a.	mhGAP (5): p. 82
Carer and family support				
Assessment of carer and family needs	Advise family members and carers about their right to the following and how to get them: · a formal assessment of their own needs (known as a 'Carer's Assessment'), including their physical and mental health) · short breaks and other respite care.	Strong	Very low	MHID (1): p. 324; 9.3
Carer and family training and support	When providing support to family members or carers (including siblings): · recognize the impact of living with or caring for a person with a learning disability and behaviour that challenges · explain how to access family advocacy · consider family support and information groups if there is a risk of behaviour that challenges, or it is emerging · consider formal support through disability-specific support groups for family members or carers and regular assessment of the extent and severity of the behaviour that challenges · provide skills training and emotional support, or information about	Strong and Conditional	Very low to moderate	CBID (2): p. 90; 5.4.1:

these, to help them take part in and support interventions for the person with a learning disability and behaviour that challenges.			
Develop a written behaviour support plan for children, young people and adults with a learning disability and behaviour that challenges that is based on a shared understanding about the function of the behaviour. This should: · identify proactive strategies designed to improve the person's quality of life and remove the conditions likely to promote behaviour that challenges, including: · identify preventive strategies to calm the person when they begin to show early signs of distress, including: · individual relaxation techniques o distraction and diversion onto activities they find enjoyable and rewarding · identify reactive strategies to manage any behaviours that are not preventable (see section 13.3), including how family members, carers or staff should respond if a person's agitation escalates and there is a significant risk of harm to them or others · incorporate risk management and take into account the effect of the behaviour support plan on the level of risk · be compatible with the abilities and resources of the person's family members, carers or staff, including managing risk, and can be implemented within these resources · be supported by data that measure the accurate implementation of the plan · be monitored using the continuous collection of objective outcome data · be reviewed frequently (fortnightly for the first 2 months and monthly thereafter), particularly if behaviour that challenges or use of restrictive interventions increases, or quality of life decreases · identify any training for family members, carers or staff to improve their understanding of behaviour that challenges shown by people with a learning disability · identify those responsible for delivering the plan and the designated person responsible for coordinating it.	Strong	Limited evidence	CBID (2): p. 171; 8.5.7
Preschool classroom-based interventions should have multiple components, including: · curriculum design and development · social and communication skills training for the children · skills training in behavioural strategies for parents or carers · training on how to mediate the intervention for preschool teachers.	Strong	Very low to low	CBID (2): p. 195; 9.4.1
Consider parent-training programs for parents or carers of children with a learning disability who are aged under 12 years with emerging, or at risk of developing, behaviour that challenges.	Conditional	Moderate	CBID (2): p. 255; 11.3.1
Consider parent training programs specifically designed for parents or carers of children with learning disabilities to help prevent or treat mental health problems in the child and to support carer wellbeing.	Conditional	Very low to moderate	MHID (1): p. 176; 5.3.5
Parent training programs should: · be delivered in groups of parents or carers · be accessible (for example, take place outside normal working hours or in community settings with childcare facilities) · focus on developing communication and social functioning skills · typically consist of 8 to 12 sessions lasting 90 minutes · follow the relevant treatment manual · use all of the necessary materials to ensure consistent implementation of the programme · seek parent feedback.	Strong	Very low to moderate	MHID (1): p. 176; 5.3.5
When providing support to family members (including siblings) and carers: · recognize the potential impact of living with or caring for a person with learning disabilities and a mental health problem · explain how to access: · family advocacy · family support and information groups · disability-specific support groups for family members or carers · provide skills training and emotional support, or information about how to access these, to help them take part in and support interventions for the person with learning disabilities and a mental health problem.	Strong	Very low	MHID (1): p. 324; 9.3
Caregiver skills training should be provided for management of children and adolescents with developmental disorders, including intellectual disabilities and pervasive developmental disorders (including autism).	Strong	Low	mhGAP (5): CHILD
Behavioural interventions for children and adolescents, and caregiver skills training, may be offered for the treatment of behavioural disorders.	Conditional	Low	mhGAP (5): CHILD
Behavioural interventions (behavioural and cognitive behavioural therapies, school-based therapies, and caregiver skills training).			

	Psychological interventions, such as cognitive behavioural therapy (CBT), interpersonal psychotherapy (IPT) for children and adolescents with emotional disorders, and caregiver skills training focused on their caregivers, may be offered for the treatment of emotional disorders.	Conditional	Low	mhGAP (5): CHILD
	Provide advice on age-appropriate stimulation and parenting.	Not reported	Not reported	mhGAP (5): CHILD
	When providing support to family members or carers (including siblings): · recognize the impact of living with or caring for a person with a learning disability and behaviour that challenges · explain how to access family advocacy · consider family support and information groups if there is a risk of behaviour that challenges, or it is emerging · consider formal support through disability-specific support groups for family members or carers and regular assessment of the extent and severity of the behaviour that challenges · provide skills training and emotional support, or information about these, to help them take part in and support interventions for the person with a learning disability and behaviour that challenges.	Strong and Conditional	Very low to moderate	CBID (2): p. 90; 5.4.1
	Provide carer support.	Not reported	Not reported	mhGAP (5): CHILD
	When providing support to family members (including siblings) and carers: · recognize the potential impact of living with or caring for a person with learning disabilities and a mental health problem · explain how to access: - family advocacy, - family support and information groups, - disability-specific support groups for family members or carers · provide skills training and emotional support, or information about how to access these, to help them take part in and support interventions for the person with learning disabilities and a mental health problem.	Strong	Very low	MHID (1): p. 324; 9.3
	When providing support to family members or carers (including siblings): · recognize the impact of living with or caring for a person with a learning disability and behaviour that challenges · explain how to access family advocacy · consider family support and information groups if there is a risk of behaviour that challenges, or it is emerging · consider formal support through disability-specific support groups for family members or carers and regular assessment of the extent and severity of the behaviour that challenges · provide skills training and emotional support, or information about these, to help them take part in and support interventions for the person with a learning disability and behaviour that challenges.	Strong and Conditional	Very low to moderate	CBID (2): p. 90; 5.4.1:
Mental health (incl. distress, anger, depression and risk to self and others)				
Assessment of mental health (incl. distress, depression, risk to self and others)	Staff should consider using identification questions (adjusted as needed) as recommended in the NICE guidelines on specific mental health problems to identify common mental health problems in people with learning disabilities.	Conditional	Weak to strong	MHID (1): p. 79; 4.4
	When conducting mental health assessments, take into account the person's: · level of distress · understanding of the problem · living arrangements and settings where they receive care · strengths and needs.	Strong	Limited evidence	MHID (1): p. 102; 4.6
	During any mental health assessment: · consider using tools that have been developed or adapted for people with learning disabilities and · take cost into account if more than one suitable tool is available.	Conditional	Limited evidence	MHID (1): p. 103; 4.6
	If a mental health problem is suspected in a person with learning disabilities, staff should conduct a triage assessment to establish an initial formulation of the problem. This should include: · a description of the problem, including its nature, severity and duration · an action plan including possible referral for further assessment and interventions.	Strong	Weak to strong	MHID (1): p. 79; 4.4
	Conduct an initial assessment for people who are experiencing a mental health crisis, which should: · include an assessment of the person's mental health · include a risk assessment (see recommendations 23-25) · include identification of interventions to: o help address the problem that caused the crisis · minimize any associated risks · bring stability to the individual and their immediate environment · produce a crisis plan that sets out (using the least restrictive options possible) how to reduce the likelihood of further crises, and what to do if the person has another crisis.	Strong	Limited evidence	MHID (1): p. 105; 4.6

	When assessing depressive symptoms in an adult with learning disabilities, consider using a formal measure of depression to monitor change over time, such as the Glasgow Depression Scale (the self-report for people with milder learning disabilities or the carer supplement for people with any degree of learning disabilities).	Conditional	Limited evidence	MHID (1): p. 104; 4.6
	When conducting risk assessments with people with learning disabilities and mental health problems, assess: · risk to self · risk to others (including sexual offending) · risk of self-neglect · vulnerability to exploitation · likelihood and severity of any particular risk · potential triggers, causal or maintaining factors · whether safeguarding protocols should be implemented.	Strong	Limited evidence	MHID (1): p. 104; 4.6
	Assess and regularly review the following areas of risk during any assessment of behaviour that challenges: · suicidal ideation, self-harm (in particular in people with depression) and self-injury · harm to others · self-neglect · breakdown of family or residential support · exploitation, abuse or neglect by others · rapid escalation of the behaviour that challenges. Ensure that the behaviour support plan includes risk management (see recommendation 33).	Strong	Limited evidence	CBID (2): p. 169; 8.5.3
	Conducting an assessment for mental, neurological, and substance use (MNS) conditions involves the following steps. First, the presenting complaint is explored, then a history is obtained including asking about past MNS issues, general health problems, family MNS history, and psychosocial history. Observe the person (Mental Status Exam), establish a differential diagnosis, and identify the MNS condition. As part of the assessment, conduct a physical examination and obtain basic laboratory tests as needed. The assessment is conducted with informed consent of the person.	n.a.	n.a.	mhGAP (5) ^d
Antidepressants	When psychosocial interventions prove ineffective, fluoxetine (but not other Selective Serotonin Reuptake Inhibitors or Tricyclic Antidepressants) may be offered in adolescents with moderate-severe depressive episode/disorder. The intervention should only be offered under supervision of a specialist.	Conditional	Very low	mhGAP (5) ^d
Psychological therapies (incl. cognitive behavioural therapy)	Psychological interventions, such as cognitive behavioural therapy (CBT), interpersonal psychotherapy for children and adolescents with emotional disorders, and caregiver skills training focused on their caregivers, may be offered for the treatment of emotional disorders.	Conditional	Low	mhGAP (5) ^d
	Consider CBT, adapted for people with learning disabilities (see the intervention adaptation methods in 31), to treat depression or subthreshold depressive symptoms in people with milder learning disabilities.	Conditional	Very low to low	MHID (1): p. 159; 5.2.7
	Psychological interventions, such as CBT, interpersonal psychotherapy for children and adolescents with emotional disorders, and caregiver skills training focused on their caregivers, may be offered for the treatment of emotional disorders.	Conditional	Low	mhGAP (5) ^d
Physical exercise training	Advice on physical activity should be encouraged as part of treatment for adults with depressive episode/disorder with inactive lifestyles. In moderate and severe depression, this intervention should be considered as adjunct to antidepressants or brief structured psychological treatments.	Conditional	Very low	mhGAP (5) ^d
Stress management training	In non-specialized health care settings, relaxation training should be considered as treatment for anxiety symptoms (in absence of depressive episode/disorder) - who are in distress or have some degree of impaired functioning.	Conditional	Very low	mhGAP (5) ^d
	Relaxation training may be considered as treatment of adults with depressive episode/disorder. In moderate and severe depression, this intervention should be considered as adjunct to antidepressants or structured brief psychological treatments.	Conditional	Very low	mhGAP (5) ^d

^a See Table D2.6 for explanation of the strength of recommendation categories. ^b See Table D2.7. for explanation of the quality of evidence categories. ^c See Table D2.5. for reference to the clinical practice guideline. ^d See mhGAP Evidence Resource Center for original recommendations.

ADL: activities of daily living; CBT: cognitive behavioural therapy; DG: development group; DID: disorders of intellectual development; MNS: mental, neurological and substance use; n.a.: not available.

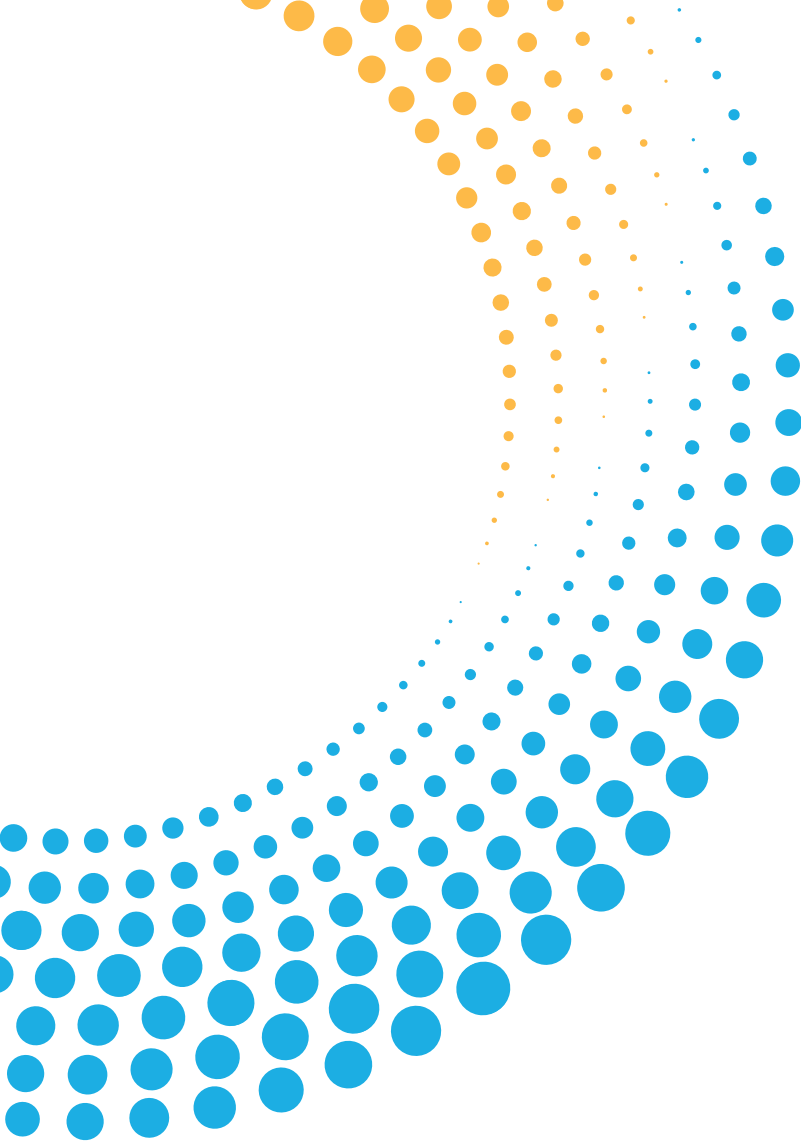
Table D2.10. Evidence table for evidence from Cochrane systematic reviews for assessments and interventions included in the *Package of interventions for rehabilitation for disorders of intellectual development*

Assessments and interventions in the PIR	Intervention	Control	Outcomes	No. studies	No. participants	Corresponding risk (95% CI)	Heterogeneity	Quality of the evidence	Reference
Problems with behaviours									
Behavioural interventions	Anger management (community setting)	Wait-list control group	Aggressive behaviour: severity of incidents	1	158	MD 1.7 (-2.55 to 5.95)	-	Moderate	Ali, 2015 (6)
			Ability to control anger	1	162	MD -6.3 (-13.53 to 0.93)	-	Moderate	
			Ability to control anger	1	161	MD -6.9 (-13.09 to -0.71)	-	Moderate	
			Mental state: depression - post-treatment	1	157	MD -0.7 (-3.06 to 1.66)	-	Moderate	
			Mental state: anxiety	1	154	MD -2.3 (-5.39 to 0.79)	-	Moderate	
			Quality of life - self reported	1	129	MD -5.6 (-18.11 to 6.91)	-	Moderate	
			Cost of service utilisation: Client Service Receipt Inventory (CSRI)	1	133	MD 102.99 (-117.16 to 323.14)	-	Moderate	
Relaxation training	Anger management (forensic setting)	Wait-list control group	Ability to control anger	1	36	MD -8.7 (-19.27 to 1.87)	-	Very low	Ali, 2015 (6)
			Ability to control anger	1	36	MD -2.06 (-5.5 to 1.38)	-	Very low	
	Meditation based on mindfulness (community setting)	Wait-list control group	Aggressive behaviour (physical aggression)	1	34	MD -2.8 (-4.37 to -1.23)	-	Low	
			Aggressive behaviour (verbal aggression)	1	34	MD -3.3 (-5.05 to -1.55)	-	Low	
	Modified relaxation training (community setting)	No-treatment control group	Aggressive behaviour (all disruptive behaviours)	1	10	MD -7.3 (n.a.)	-	Very low	
			Aggressive behaviour (verbal disruptive behaviours)	1	10	MD -2.9 (n.a.)	-	Very low	

CI: confidence interval; MD: mean difference; No: number; PIR: package of interventions for rehabilitation.

2.5. References

1. Mental health problems in people with learning disabilities: prevention, assessment, and management (NG 54). London: National Institute for Health and Care Excellence (NICE); 2016 (<https://www.nice.org.uk/guidance/ng54/resources/mental-health-problems-in-people-with-learning-disabilities-prevention-assessment-and-management-pdf-1837513295557>, accessed February 2023).
2. Challenging behaviour and learning disabilities: Prevention and interventions for people with learning disabilities whose behaviour challenges (NG 11). London: National Institute for Health and Care Excellence (NICE); 2015 (<https://www.nice.org.uk/guidance/ng11/resources/challenging-behaviour-and-learning-disabilities-prevention-and-interventions-for-people-with-learning-disabilities-whose-behaviour-challenges-1837266392005>, accessed February 2023).
3. Sullivan W, Diepstra H, Heng J, Ally S, Bradley E, Casson I, Hennen B, et al. Primary care of adults with intellectual and developmental disabilities: 2018 Canadian consensus guidelines. *Can Fam Physician* 2018; 64(4):254–79.
4. Appraisal of Guidelines for Research & Evaluation. The AGREE Next Steps Consortium. The AGREE II Instrument. Update 2017. The Agree Research Trust 2009. (<https://www.agreetrust.org/wp-content/uploads/2017/12/AGREE-II-Users-Manual-and-23-item-Instrument-2009-Update-2017.pdf>, accessed 8 June 2023).
5. mhGAP Intervention guide for mental, neurological and substance use disorders in non-specialized health settings. Version 2.0, Geneva: World Health Organization; 2016 (<https://apps.who.int/iris/handle/10665/250239>, accessed October 2022).
6. Ali A, Hall I, Blickwedel J, Hassiotis A. Behavioural and cognitive-behavioural interventions for outwardly directed aggressive behaviour in people with intellectual disabilities. *Cochrane Database Syst Rev*. 2015(4):CD003406.
7. Torra Moreno M, Canals Sans J, Colomina Fosch M. Behavioral and cognitive interventions with digital devices in subjects with intellectual disability: a systematic review. *Front Psychiatry*. 2021;12:647399.
8. Engel-Yeger B, Hardal-Nasser R, Gal E. Sensory processing dysfunctions as expressed among children with different severities of intellectual developmental disabilities. *Res Dev Disabil*. 2011;32(5):1770–5.
9. Vuijk P, Hartman E, Scherder E, Visscher C. Motor performance of children with mild intellectual disability and borderline intellectual functioning. *J Intellect Disabil Res*. 2010;54(11):955–65.
10. Hocking J, McNeil J, Campbell J. Physical therapy interventions for gross motor skills in people with an intellectual disability aged 6 years and over: a systematic review. *Int J Evid Based Healthc*. 2016;14(4):166–74.
11. Kapsal N, Dicke T, Morin A, Vasconcellos D, Maïano C, Lee J, et al. Effects of physical activity on the physical and psychosocial health of youth with intellectual disabilities: a systematic review and meta-analysis. *J Phys Act Health*. 2019;16(12):1187–95.
12. Burns C, Lemon J, Granpeesheh D, Dixon D. Interventions for daily living skills in individuals with intellectual disability: a 50-year systematic review. *Adv Neurodev Disord*. 2019;3:235–45.



9789240071216



9 789240 071216